Copernicus Global Land Operations
”Vegetation and Energy”
“CGLOPS-1”

Framework Service Contract N° 945120-IPR-2023 (JRC)

ALGORITHM THEORETHICAL BASIS DOCUMENT

Leaf Area Index (LAI)
Fraction of Absorbed Photosynthetically Active Radiation (FAPAR)
Fraction of green Vegetation Cover (FCover)
Version 2.0

Issue I1.00

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List of Acronyms

AD	Applicable Document
ATBD	Algorithm theoretical based Document
AATSR	Advanced Along Track Scanning Radiometer
BELMANIP	BEnchmark Land Multisite ANalysis and Intercomparison of Products
C6	Collection 6
CCI-LC	Climate Change Initiative-Land Cover
CEOS	Committee for Earth Observation Satellite
CG	Crops and Grassland
CLMS	Copernicus Land Monitoring Service
COG	Cloud-Optimized Geotiff
CYCLOPES	Carbon cYcle and Change in Land Observational Products from an Ensemble of Satellites
DBF	Deciduous Broadleaf Forest
DN	Digital Number
EBF	Evergreen Broadleaf Forest
FAPAR	Fraction of Absorbed Photosynthetically Active Radiation
FCover	Fraction of vegetation cover
FIPAR	Fraction of Intercepted Photosynthetically Active Radiation
FP5/FP7	5th and 7th Framework Programme
JRC	Joint Research Center
GAUL	Global Administrative Unit Layer
GBOV	Ground-Based Observations for Validation
GCOS	Global Climate Observation System
GEOCLIM	Climatology of Version1 LAI, FAPAR, Fcover VGT products
GLASS	Global Land Surface Satellite
GLOBCOVER	Global land cover from MERIS
GMES	Global Monitoring of Environment and Security
IDEPIX	Identification of Pixels processor
ImagineS	Implementation of Multi-scale Agricultural Indicators Exploiting Sentinels
LAI	Leaf Area Index

LSA SAF	Land Surface Analysis Satellite Applications Facility
MERIS	Medium Resolution Imaging Spectrometer
MODIS	Moderate Imaging Spectrometer
NDVI	Normalized Difference Vegetation Index
NF	Needleleaf Forest
NIR	Near Infrared spectral domain
NNT	Neural Network Technique
NOBS	Number of observations
OLCI	Ocean and Land Colour Instrument
PROBA-V	Project of on-board autonomy – VEGETATION instrument
PUM	Product User Manual
PV	PROBA-V
QA	Quantitative quality assessment
QAR	Quality Assessment Report
QC	Qualitative quality index
R&D	Research and Development
RMSE	Root Mean Square Error
RT	Real Time
S3	Sentinel-3
SAA	Sun Azimuth Angle
SLSTR	Sea and Land Surface Temperature Radiometer
SMAC	Simplified Method for Atmospheric Correction
SPOT	Satellite Pour l'Observation de la Terre
SSB	Shrubs/Savana/Bare soil
SSD	Service Specifications Document
SVP	Service Validation Plan
SZA	Sun Zenith Angle
TOA	Top of Atmosphere
TOC	Top of Canopy
VAA	Viewing Azimuth Angle
VGT	VEGETATION instrument onboard SPOT satellite

VZA	Viewing Zenith Angle
WGS84	World Geodetic System 1984

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EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified bio-geophysical products on the status and evolution of the land surface. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. Production and delivery of the parameters take place in a timely manner and are complemented by the constitution of long-term time series.

From 1st January 2013, the Copernicus Land Monitoring Service is providing Essential Climate Variables like the Leaf Area Index (LAI), the Fraction of Absorbed Photosynthetically Active Radiation absorbed by the vegetation (FAPAR), as well as the Fraction of Green Vegetation Cover (FCover), every 10 days over the globe, on a reliable and automatic basis from Earth Observation satellite data.

The operations of the CLMS are supported by a number of research and development initiatives. Among them, the FP7 ImagineS project (http://FP7-imagines.eu) set-up the Version 1.0 of the algorithm retrieving the LAI, FAPAR, FCover 300m products from PROBA-V data The Version 1.0 of these products, produced and delivered in near Real Time (RT), were derived from PROBA-V Collection 1 data from January 2014 to June 2020, end of the operational PROBA-V mission.. The Version 1.1. of the algorithm was adapted to the imagery of the Ocean and Land Colour Instrument (OLCI) onboard the Sentinel-3 platform. The Version 1.1 Sentinel-3 products were delivered from July 2020 ensuring the continuity of PROBA-V LAI, FAPAR, and FCover 300m products. The quality assessment was reported in [CGLOPS1_QAR_LAI[FAPAR/FCOVER]300m-V1.1]. Some marginal inconsistencies were identified between PROBA-V and Sentinel-3 300m products due to the differences between Version 1.0 and Version 1.1 retrieval algorithms and input auxiliary data, and because of some issues in the implementation of Version 1.0 and Version 1.1 algorithms. Further, the Collection 2 of PROBA-V data has been released in March 2023. To improve the consistency of the long time series of LAI, FAPAR and FCover products at 300m resolution from 2014 to present, we have adapted Version 1.1 to the characteristics of PROBA-V Collection 2 input data and to the Version 2.3 of Sentinel-3 OLCI surface reflectance for the reprocessing of the entire time series, from PROBA-V and Sentinel-3, resulting in Version 2.0 algorithm.

This Algorithm Theoretical Based Document (ATBD) describes the Version 2.0 of the algorithm for the generation of LAI, FAPAR and FCover 300m products from PROBA-V Collection 2 (2014-2018) and Sentinel-3 OLCI (2019-present) data. The variables are calculated globally on a 10-daily basis and made available to the user in near-real time every 10 days. The product is projected on a regular latitude/longitude grid with a resolution of 1/336°. It is delivered covering the whole globe (from -180°E to +180°W and from +85°N to -65°S) and provided in netCDF4-CF format (by default format) containing the variables values (LAI, FAPAR and FCover), associated with some quantitative and qualitative quality indicators.

1 BACKGROUND OF THE DOCUMENT

1.1 SCOPE AND OBJECTIVES

CLMS supports applications in a variety of domains such as spatial and urban planning, forest management, water management, agriculture and food security, nature conservation and restoration, rural development, ecosystem accounting and mitigation/adaptation to climate change. The products are then operationally generated and delivered freely in near real time through the CLMS portal (https://land.copernicus.eu).

This document provides a detailed description and justification of the algorithm proposed for Version 2.0 of the algorithm of the LAI, FAPAR and FCover 300m derived from PROBA-V and Sentinel-3 Top-Of-Canopy (TOC) reflectance.

A theoretical validation is done including a comparison with Version 1.0 of PROBA-V 300m products (Fuster et al. 2020) and MODIS Collection 6 products (Myneni et al. 2002; Yan et al. 2016a; Yan et al. 2016b). Further validation of the products is completed by a full quality assessment analysis according to the Product Quality Assurance Document [CGLOPS1_PQAD].

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 recalls the requirements
- Chapter 3 contains the definition of the proposed products, a description of the input data and the outline of the algorithm
- Chapter 4 describes in detail the algorithm
- Chapter 5 presents the algorithm performance

1.3 RELATED DOCUMENTS

1.3.1 Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

Available at https://etendering.ted.europa.eu/cft/cft-display.html?cftId=13795

1.3.2 Input

Document ID	Descriptor
CGLOPS1_PQAD	Product Quality Assurance Document of the Copernicus Land Monitoring Service for "Vegetation and Energy" products
CGLOPS1_ATBD_S3-AC-V1	Algorithm Theoretical Basis Document of the atmospheric corrections applied on the Sentinel-3 data
CGLOPS1_QAR_S3-CloudMask	Report presenting the evaluation of Sentinel-3 OLCI and SLSTR cloud, cloud shadow and snow masks.
CGLOPS1_PUM_S3-TOC-Reflectance300m-V2.3	Product User Manual of Sentinel-3 OLCI and SLSTR Top-Of-Canopy Reflectance Version 2.3
CGLOPS1_QAR_LAI[FAPAR/FCOVER] 300m-V1	Quality assessment report of the LAI, FAPAR, FCover 300m Version 1.0 PROBA-V products
CGLOPS1_QAR_LAI[FAPAR/FCOVER] 300m-V1.1	Quality assessment report of the LAI, FAPAR, FCover 300m Version 1.1 Sentinel-3 products

These documents are available on the CLMS website, in the Technical Library: https://land.copernicus.eu/en/technical-library

1.3.3 Output

CGLOPS1_PUM_LAI[FAPAR/FCOVER] 300m-V2.0	Product User Manual summarizing all information about LAI, FAPAR, FCover 300m Version 2.0 product
CGLOPS1_VR_LAI[FAPAR/FCOVER]300m-V2.0	Validation Report of the LAI, FAPAR, FCover 300m Version 2.0 product

These documents are available on https://land.copernicus.eu/en/technical-library

1.3.4 External documents

Document ID	Descriptor
ImagineS_RP2.1_ATBD-LAI300m_I1.73	Algorithm Theoretical Basis Document describing the retrieval methodology of the Version 1.0 of LAI, FAPAR, FCover 300m from PROBA-V, set-up in the context of the FP7/ImagineS project.

Available on: https://land.copernicus.eu/en/technical-library

PROBAV_PUM_C2 PROBA-V Collection 2 Products User Manual v1.0, 27 March 2023

Available at https://proba-v.vgt.vito.be/sites/probavvgt/files/downloads/PROBA-V_C2_Products_User_Manual.pdf

ACRI-ST, 2017 Product Data Format Specification – OLCI Level 1 products, available at:

https://sentinels.copernicus.eu/documents/247904/4812102/S3IPF PDS 004.1 - i2r5 – Product Data Format Specification – OLCI Level 1.pdf

GCOS#245 The 2022 GCOS ECVs Requirements

Available online at : https://library.wmo.int/records/item/58111-the-2022-gcos-ecvs-requirements-gcos-245?offset=1

2 REQUIREMENTS

According to the applicable document [AD1], the requirements relevant for LAI, FAPAR, FCover products are described below.

2.1 SPECIFIC TECHNICAL DETAILS AND REQUIREMENTS

PRODUCT SPECIFICATION	
Geometric Properties:	
Baseline dataset pixel resolution	300m
Target baseline location accuracy	Better than 0.5 pixels
Coordinate position	Centre of the pixel
Geodetical datum	WGS84
Geographic projection	Regular latitude/longitude grid
Geographic coverage:	Global
Temporal resolution	10-day period (dekad: days 1-10, 11-20, 21 end of month)
Timeliness	Within 2 days (optimally 1 day) after the end of each dekad

FAPAR – FRACTION OF ABSORBED PHOTOSYNTHETICALLY ACTIVE RADIATION	
FAPAR corresponds to the fraction of photosynthetically active radiation (PAR, i.e. solar radiation in the spectral range 400-700 nm) that is absorbed by plants.	
Uncertainty (2-sigma)	Threshold: 10% for values >0.5; 0.005 (absolute values) for smaller values Goal: 5% for values ≥ 0.05; 0.0025 (absolute value) for smaller values
Stability (per decade i.e. 10 years)	Threshold: 3% Goal: 1.5%

LAI – LEAF AREA INDEX	
Leaf Area Index of a plant canopy or ecosystem is defined as one half of the total green leaf area per unit horizontal ground surface area and measures the area of leaf material present in the specified environment (projection to the underlying ground along the normal to the slope) [m^2/m^2].	
Uncertainty (2-sigma)	Threshold: 20% for values ≥ 0.5; 0.1 (absolute value) for smaller values Goal: 10% for values ≥ 0.5; 0.05 (absolute value) for smaller values
Stability (per decade, i.e. 10 years)	Threshold: 3% Goal: 1.5%

FCOVER – FRACTION OF GREEN VEGETATION COVER	
The Fraction of Vegetation Cover corresponds to the fraction of ground covered by green vegetation. Practically, it quantifies the spatial extent of the vegetation.	
Uncertainty (2-sigma)	Threshold: 10% Goal: 5%
Stability (per decade, i.e. 10 years)	Threshold: 5% Goal: 2.5%

2.2 FURTHER REQUIREMENTS

2.2.1 Output product composition

Products may contain various information layers and ancillary information – the base reference for product packages are the operational products as on 01.03.2023.

2.2.2 Data structure

Data coding[1] shall be compatible with the Global Land products as on 01.03.2023 and/or follow the INSPIRE specifications, where applicable.
Ancillary information shall be as currently used and include at least the following:

- The number of measurements per pixel used to generate any synthesis product
- The per-pixel date of the individual measurement or the start-end dates of the period covered
- Quality indicators, with explicit per-pixel identification of the cause of anomalous parameter result.

The product naming and filename conventions that are used in the Copernicus Global Land component production as on 01.03.2023 shall be followed. This may be adapted for complete product collections upon agreement with the contracting authority during the Framework.

2.2.3 Data format

To ensure interoperability with the current Global Land component (operational product data formats and archive data formats) and other Copernicus services, all datasets will be available in NETCDF. Additional format such as Cloud Optimized Geotiff (COG) or ZAR format can be proposed for production or could be requested by the Contracting Authority during the Framework Contract.

[1] Data coding is the provision for the number of bits, range of values, usage of reserved values, content of status map, conversion to physical values formulas.

2.2.4 Uncertainties and validation

Uncertainties indicated in the product specifications above follow the threshold and goal proposed by GCOS#245.

Uncertainties estimates should account for the error propagation uncertainty coming from input data though the retrieval algorithms as during the contract period, ESA plans to imbed uncertainties in the Sentinel-3 ground segment products and, for Sentinel-2, there is an offline tool to determine Level 1 uncertainties; these can be used for propagation in the production chain.

Validation of the products shall conform to at least the CEOS LPV standards. Wherever appropriate the bio-geophysical variables shall be validated and compared to CEOS CAL/VAL data sets and/or Ground-Based Observations for Validation (GBOV) of Copernicus Global Land component of Copernicus Global Land component when biophysical parameters are available.

2.2.5 Input data

Copernicus sentinel data are available from https://dataspace.copernicus.eu

Bio-geophysical variable products should be based on common base reflectance data:

- Sentinel 3: Derived Top-of-Canopy reflectance may be brokered or produced as it is the case on 01.03.2023 under the CGLOPS contracts;
- Sentinel 2: the Global Land Sentinel-2 Global Mosaic (S2GM)[2] component provides temporal mosaic of surface spectral bands that can be brokered and/or directly used.

Up to and including 2019, products archives of the Global Land Component have been based on SPOT VGT, Proba-V, ENVISAT, MODIS, TOPEX/Poseidon, Jason-1, Jason-2, Jason-3, datasets, which are available through the https://dataspace.copernicus.eu.

LST and SWI can be based on geostationary and other satellite data.

Ancillary satellite data that is purchased through Copernicus and put at disposal of the Services is available through the Data Warehouse and will become available on the Copernicus Dataspace Ecosystem.

Ancillary data sets, other than satellite imagery described above, that might be required shall be the responsibility of the contractor.

[2] https://land.copernicus.eu/en/products/global-image-mosaic

2.2.6 Product delivery

Products shall be delivered to the Copernicus Land Component dissemination. The Copernicus Data Space Ecosystem infrastructure will be used to provide access to the final map products.

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3 OVERVIEW

3.1 THE CONSIDERED PRODUCTS

The considered products correspond to actual vegetation biophysical variables that are defined below.

3.1.1 Leaf Area Index (LAI)

LAI is defined as one half the total green (i.e., photosynthetically active) leaf area per unit horizontal ground surface area (Chen and Black 1992). LAI is a non-dimensional quantity, although units of m^2/m^2 are often quoted. It determines the size of the interface for exchange of energy (including radiation) and mass between the canopy and the atmosphere. This is an intrinsic canopy primary variable that should not depend on observation conditions. LAI is strongly non-linearly related to reflectance. Therefore, its estimation from remote sensing observations is scale dependent (Garrigues et al. 2006; Weiss et al. 2000). Note that vegetation LAI as estimated from remote sensing includes all the green contributors such as the understory when existing under forests canopies.

3.1.2 FAPAR

The FAPAR is defined as the fraction of photosynthetically active radiation (PAR; solar radiation reaching the surface in the 0.4-0.7µm spectral region) that is absorbed by vegetation. FAPAR is expressed as a unitless fraction of the incoming radiation received at the land surface. The FAPAR value results directly from the radiative transfer in the canopy. FAPAR is the sum of two terms, weighted by the diffuse fraction in the PAR domain: the 'black sky' FAPAR that corresponds to the direct component (collimated beam irradiance in the sun direction only) and the 'white sky' or the diffuse component. FAPAR can be computed at a given time (e.g instantaneous FAPAR at the actual sun position of measurement) or daily integrated.

FAPAR depends on canopy structure, vegetation element optical properties and illumination conditions (Baret et al. 2007). It is very useful as input to a number of primary productivity models based on simple efficiency considerations (McCallum et al. 2009; Prince 1991). Most of the primary productivity models using this efficiency concept are running at the daily time step. Since the CLMS FAPAR product is originally derived from CYCLOPES FAPAR at 10:00 (Baret et al. 2007) and MODIS aboard Terra FAPAR at 10:30 (Myneni et al. 2002), CLMS FAPAR product is defined as the black-sky fraction of PAR absorbed by green elements including over and understorey vegetation at 10:15 (actually between 10:00 and 10:30) which is a good approximation of the daily integrated black-sky FAPAR value (Baret et al. 2007).

FAPAR is relatively linearly related to reflectance values, and is little sensitive to scaling issues (Hilker et al. 2010; Weiss et al. 2000). Note also that the FAPAR refers only to the green parts of the canopy.

3.1.3 Fraction of green Vegetation Cover (FCover)

FCover is defined as the fraction of ground surface covered by green vegetation as seen from the nadir direction (Baret et al. 2013). This definition agrees with the definition of LSA SAF FCover products (https://landsaf.ipma.pt/en/products/vegetation/fvc/) as well as the GLASS FCover product (Jia et al. 2015). FCover is expressed as a unitless fraction of the ground surface.

FCover is used to separate vegetation and soil in energy balance processes, including temperature and evapotranspiration. It is computed from the leaf area index and other canopy structural variables and does not depend on variables such as the geometry of illumination as compared to FAPAR. For this reason, it is a very good candidate for the replacement of classical vegetation indices for the monitoring of green vegetation. Because of its quasi-linear relationship with reflectances, FCover is only marginally scale dependent (Weiss et al. 2000). Note that similarly to LAI and FAPAR, only the green elements are considered.

3.2 PROBA-V INSTRUMENTS AND DATA

The PROBA-V Collection 2 (C2) daily synthesis (S1) top of canopy (TOC) reflectance products [PROBAV_PUM_C2] have been used in Version 2.0 of 300m algorithm.

The PROBA-V sensor has been launched on 6th May 2013 onboard the PROBA platform. It was designed to bridge the gap in space-borne vegetation measurements between SPOT-VGT (March 2018 – May 2014) and the Sentinel-3 satellites launched in 2016. The mission objective is to ensure the continuity with the heritage of the SPOT-VGT mission.

PROBA-V operated at an altitude of 820 km altitude in a sun-synchronous orbit with a local overpass time at launch of 10:45 h. Because the satellite had no onboard propellant, the overpass time was expected to gradually differ from the at-launch value. After launch, the local overpass time first increased to 10:50h in October 2014, followed by a decrease to 10:45h in June 2016. By end-of-mission in June 2020, the Local Time of Descending Node will be at ~09:30h.

The instrument had a Field of View of 102°, resulting in a swath width of 2295 km. This swath width ensured a daily near-global coverage (90%) and full global coverage is achieved every 2 days. An array of 6000x4 elements was used in the VIS-NIR (only 3 bands on the 4 potential ones are used) yielding to a ground sampling distance that varied across the swath from 100m up to 350m at the extremities of the swath (Figure 1, left). The SWIR domain was sampled using 3 arrays of 1024 elements, providing a ground sampling distance about twice as that in the VIS-NIR (Figure 1, right).

This obviously posed a problem regarding the consistency of the radiometric information between the VIS-NIR and SWIR domains.

The optical design of PROBA-V consisted of three cameras. Each camera had two focal planes, one for the short-wave infrared (SWIR) and one for the visible and near-infrared (VNIR) bands. The VNIR detector consisted of four lines of 5200 pixels. Three spectral bands were implemented, comparable with SPOT-VGT: BLUE, RED, and NIR (see Table 1). The SWIR detector was a linear array composed of three staggered detectors of 1024 pixels. The normalized spectral response functions of the four spectral bands of PROBA-V are shown in Figure 2.

The PROBA-V processing was described in Sterckx et al. (2014) and Dierckx et al. (2014). Information on the PROBA-V Collection 2 is available on https://proba-v.vgt.vito.be/en/quality/product-and-algorithm-information. The description of the PROBA-V S1 TOC products is summarized in Table 2 and detailed in the Product User Manual [PROBAV_PUM_C2].

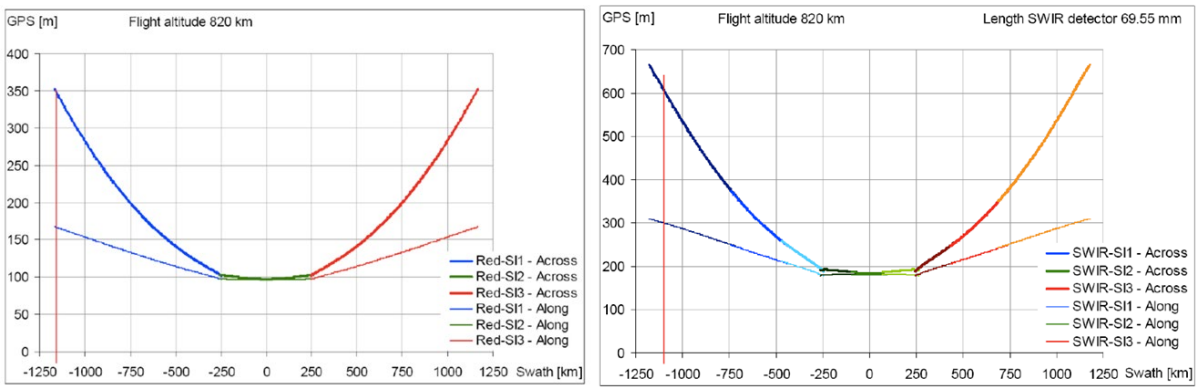


Figure 1: Ground sampling distance (GPS in m) as a function of the position on the swath (in km) for the VIS-NIR (left) and SWIR bands (right).

Table 1: PROBA-V spectral characteristics: band center and width. The spectral bands selected for Version 2.0 of LAI, FAPAR, FCover 300m algorithm are highlighted in bold.

Acronym	Center (nm)	Width (nm)	Potential Applications
B0 (blue)	**463**	**46**	**Continental ecosystems - Atmosphere**
B2 (red)	**655**	**79**	**Continental ecosystems**
B3 (NIR)	**845**	**144**	**Continental ecosystems**
SWIR	1600	73	Continental ecosystems

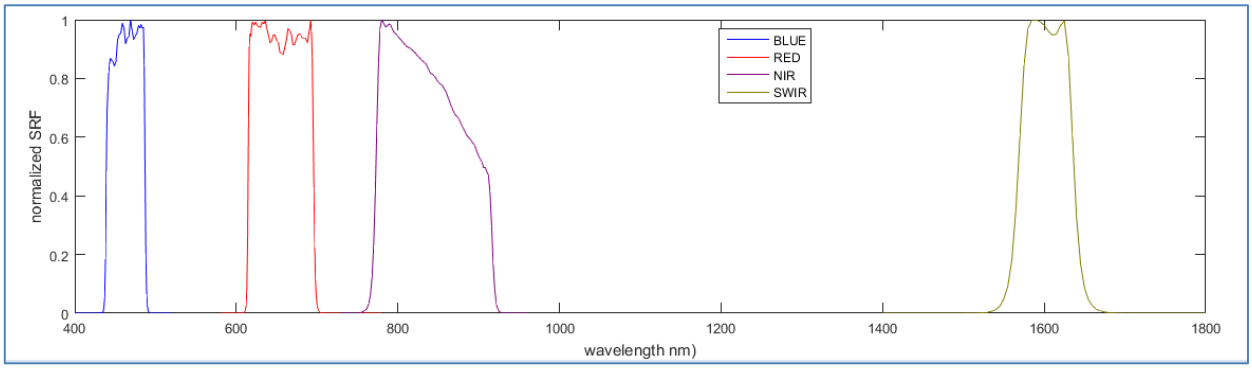


Figure 2: PROBA-V normalized spectral response function.

Table 2: PROBA-V S1 data descriptor

PROBA-V planes	Description
B0	B0 spectral band, Radiometry data
B2	B2 spectral band, Radiometry data
B3	B3 spectral band, Radiometry data
SWIR	SWIR spectral band, Radiometry data
NDVI	Normalized Difference Vegetation Index data
QC	Quality Control Bit NR 7: Radiometric quality for B0 coded as 0 if bad and 1 if good Bit NR 6: Radiometric quality for B2 coded as 0 if bad and 1 if good Bit NR 5: Radiometric quality for B3 coded as 0 if bad and 1 if good Bit NR 4: Radiometric quality for MIR coded as 0 if bad and 1 if good Bit NR 3: land code 1 or water code 0 Bit 2: snow/ice code 1 or code 0 if no ice/snow Bit 1: 0 0 1 1 0 Bit 0: 0 1 0 1 0 Clear Shadow Undefined Cloud Ice
VZA-VNIR	view zenith angles for Visible and Near Infra-Red channels
VAA-VNIR	view azimuth angles for Visible and Near Infra-Red channels
VZA-SWIR	View zenith angles for SWIR channel
VAA-SWIR	View azimuth angles for SWIR channel
SZA	sun zenith angles
SAA	sun azimuth angles
TIME	Observation timing information

3.3 SENTINEL-3 INSTRUMENTS AND DATA

Sentinel-3 is a constellation of at least 2 identical satellites, Sentinel-3A and 3B. Sentinel-3A was launched on 16 February 2016 with 4 instruments on board (Figure 3) followed by the launch of Sentinel-3B on 25 April 2018. Each satellite operates in a reference orbit with a repeat cycle of 27 days, with a 4-day sub-cycle. The use of Sentinel-3A and Sentinel-3B satellites in conjunction enable a short revisit time of less than two days at the equator.

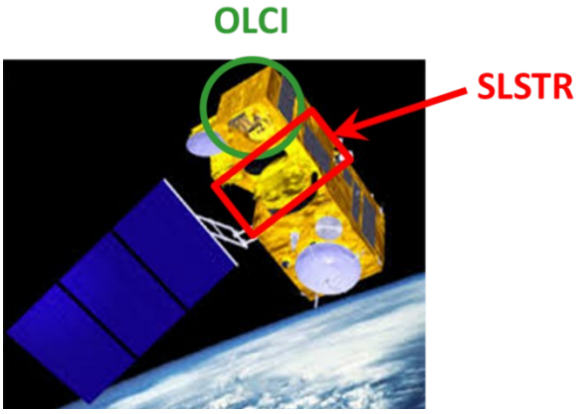


Figure 3: The Sentinel-3 platform with the Ocean and Land color Imager (OLCI) and Sea and Land Surface Temperature Radiometer (SLSTR) instruments.

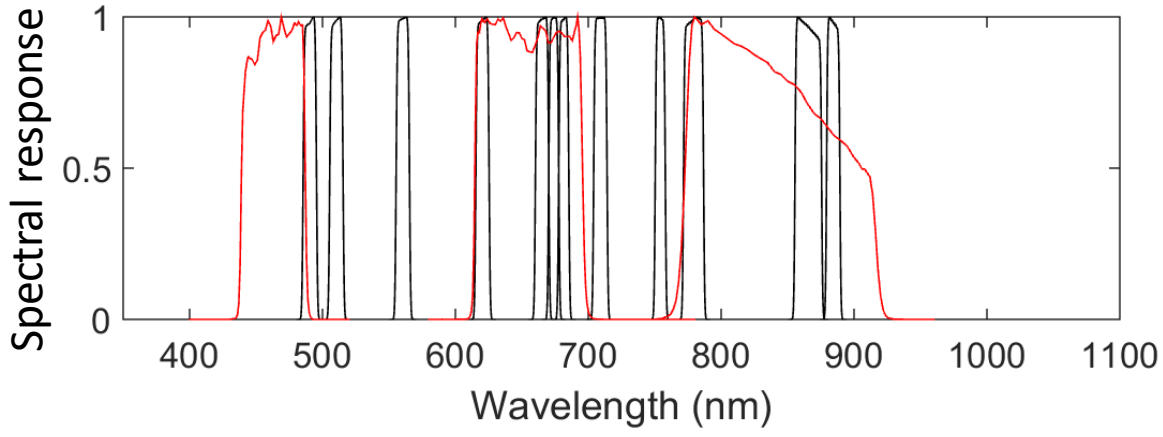


Figure 4: Spectral response of PROBA-V (red) and OLCI (black) selected bands (Oa4-12, Oa16-18; Table 3) for LAI, FAPAR, FCover 300m products.

Table 3: Sentinel-3 OLCI spectral characteristics: band center and width. The spectral bands selected for LAI, FAPAR, FCover 300m algorithm are highlighted in bold.

Band	λ centre (nm)	Width (nm)	Potential Application
Oa1	400	15	Aerosol correction, improved water constituent retrieval
Oa2	412.5	10	Yellow substance and detrital pigments (turbidity)
Oa3	442.5	10	Chlorophyll absorption max., biogeochemistry, vegetation
Oa4	**490**	**10**	**High Chlorophyll, other pigments**
Oa5	**510**	**10**	**Chlorophyll, sediment, turbidity, red tide**
Oa6	**560**	**10**	**Chlorophyll reference (Chlorophyll minimum)**
Oa7	**620**	**10**	**Sediment loading**
Oa8	**665**	**10**	**Chlorophyll (2nd Chlorophyll abs. max.), sediment, yellow substance /vegetation**
Oa9	**673.75**	**7.5**	**For improved fluorescence retrieval and to better account for smile together with the bands 665 and 680 nm**
Oa10	**681.25**	**7.5**	**Chlorophyll fluorescence peak, red edge**
Oa11	**708.75**	**10**	**Chlorophyll fluorescence baseline, red edge transition**
Oa12	**753.75**	**7.5**	**Oxygen absorption/clouds, vegetation**
Oa13	761.25	2.5	Oxygen absorption band/aerosol correction
Oa14	764.375	3.75	Atmospheric correction
Oa15	767.5	2.5	Oxygen absorption band used for cloud top pressure, fluorescence over land
Oa16	**778.75**	**15**	**Atmospheric correction/aerosol correction**
Oa17	**865**	**20**	**Atmospheric correction/aerosol correction, clouds, pixel co-registration**
Oa18	**885**	**10**	**Water vapour absorption reference band. Common reference**
Oa19	900	10	Water vapour absorption, vegetation monitoring
Oa20	940	20	Water vapour absorption, atmos./aerosol corr.
Oa21	1 020	40	Atmospheric correction/aerosol correction

The satellite flies at 814 km altitude on a circular sun-synchronous orbit with 10:00 am equatorial crossing time. The two optical instruments, OLCI (300m resolution) and SLSTR (500m resolution for optical bands and 1km for thermal ones) provide a common quasi-simultaneous view of the Earth. The SLSTR data is not used due to its lower spatial resolution. The OLCI has a swath of 1269 km, a field of view of 68° and 21 spectral channels (Table 3). The field-of-view is divided between five cameras on a common structure with the calibration assembly. Each camera has an optical grating to provide the minimum baseline of 16 spectral bands required by the mission together with the potential for optional bands for improved atmospheric corrections (Table 3 and Figure 4). Although

preliminary designed for ocean applications, OLCI is fully suitable for land applications, considering its spectral characteristics and spatial resolution.

The instantaneous top of canopy (TOC) reflectance products [CGLOPS1_PUM_S3-TOC-Reflectance300m-V2.3] from OLCI in 12 spectral bands (Oa4-12 and Oa16-18, Figure 4 and Table 3) have been used in LAI, FAPAR, FCover 300m algorithm.

3.4 RATIONALE FOR THE ALGORITHM SELECTION AND DESIGN

The objective is to develop an algorithm dedicated to the estimation of Version 2.0 of LAI, FAPAR and FCover 300m from the PROBA-V (2014-2018) and Sentinel-3 (2019-present) series of observations. The algorithm should provide high level of consistency with Version 1.0 of PROBA-V products. The Version 2.0 of LAI, FAPAR and FCover 300m products should have the same temporal sampling frequency of 10 days. Products should also be associated with quality assessment flags as well as quantified uncertainties. The algorithm runs at the pixel level without interactions with the surrounding pixels. The algorithm should provide real time estimation. This forces to perform short term projection of the product dynamics.

Version 1.0 of 300m algorithm was set-up in the ImagineS project [ImagineS_RP2.1_ATBD-LAI300m_I1.73]. After the quality assessment of the 300m LAI, FAPAR, FCover Version 1.0 [CGLOPS1_QAR_LAI[FAPAR/FCOVER]300m-V1], the methodology has been adapted to reduce the noise in the product time series especially in near real time mode and to address external recommendations and operational constraints in CLMS. This ATBD describes the Version 2.0 of the retrieval algorithm. Since the performance of LAI, FAPAR, FCover estimates is highly dependent on the level of noise in the input reflectance data, Top of Atmosphere (TOA) reflectance used in Version 1.0 have been replaced by TOC reflectance in the Version 2.0 of the algorithm. Further, the parametrization of the algorithm has been adapted in Version 2.0 to the characteristics of PROBA-V Collection 2 and Sentinel-3 Version 2.3 TOC reflectances. The neural networks used to transform PROBA-V and Sentinel-3 TOC reflectance into instantaneous LAI, FAPAR and FCover estimates have been calibrated with current Version 1.0 PROBA-V 300m LAI, FAPAR, FCover products to ensure a good agreement and temporal consistency in the transition from PROBA-V to Sentinel-3. Details of the Version 2.0 of LAI, FAPAR and FCover 300m are provided in Chapter 4. The differences with previous versions are detailed in Section 4.5.

3.5 ALGORITHM OUTLINE

The scheme proposed for the 300m products retrieval methodology is sketched in Figure 5. The algorithm starts from instantaneous TOC reflectance products which are first transformed into

instantaneous estimates of LAI, FAPAR, FCover (Step A in Figure 5). This Step A is sensor specific. Then, smoothing and gap filling is achieved over a compositing temporal window that may be dissymmetric as in the case of the near-real time situation or at the beginning of the time series (Step B in Figure 5).

Two different processing chains are applied whether the pixel's biome is an Evergreen Broadleaf Forests (EBFs) or not to account the specific behaviors of EBFs: (i) lower near infrared reflectance level for a given LAI value mainly due to thicker leaves, and (ii) high level of noise in the time series due to the remaining atmospheric effects and cloud contamination in regions characterized by a high occurrence of clouds. Gap filling is only applied for EBFs.

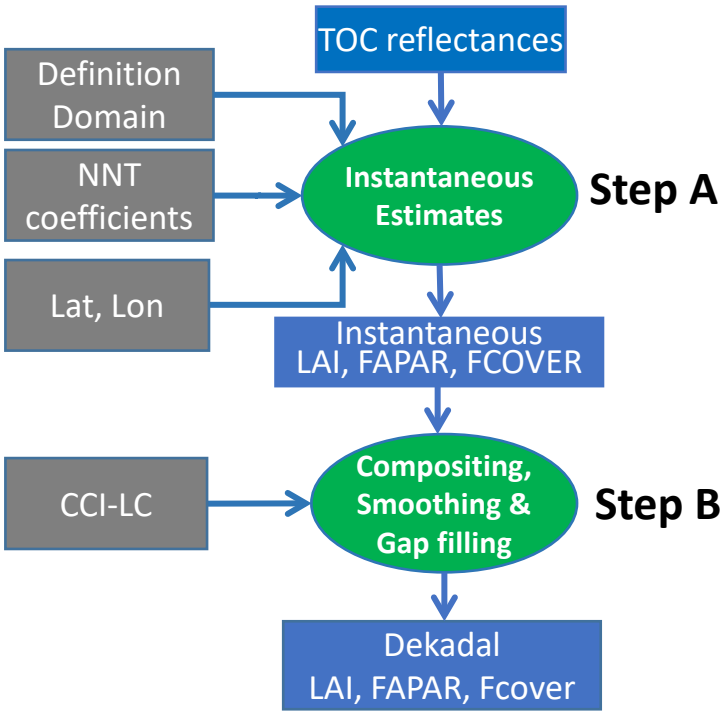


Figure 5: Flow chart showing the two processing steps in Version 2.0 LAI, FAPAR, FCover 300m algorithm.

The selection of the instantaneous estimates considered as valid in the compositing window (Step B) is made based on the LAI product.

The compositing for real time estimates (Step B) is achieved according to the scheme described in Figure 6. We consider a particular dekadal date, *D*.

- In real time mode, RT0 (when the actual date corresponds to *D*, i. e. the first line in Figure 6), the compositing is achieved using only the past observations, with a maximum compositing window spanning *c*- days in the past (*c*- was set to 210 days for EBF and 60 days for nonEBF).

The value of the product at dekadal date *D* is therefore relatively unstable in the presence of noisy and missing data.

- Then, the observations accumulate after dekadal date *D* when time passing. The value at dekadal date *D* is updated using the observations available after dekadal date *D* within the consolidation period. The accuracy of the product value progressively improves during this consolidation period.
- At the end of the consolidation period, the value of the product at dekadal date *D* has converged towards the 'historical' time series when, for a given dekad 'D' to be processed, the 'n' dekads before 'D' and after 'D' in the time series are available. 'n' is the number of dekads required for the convergence of LAI, FAPAR and FCover values. 'n' was fixed to 6 dekads, i.e. 60 days.

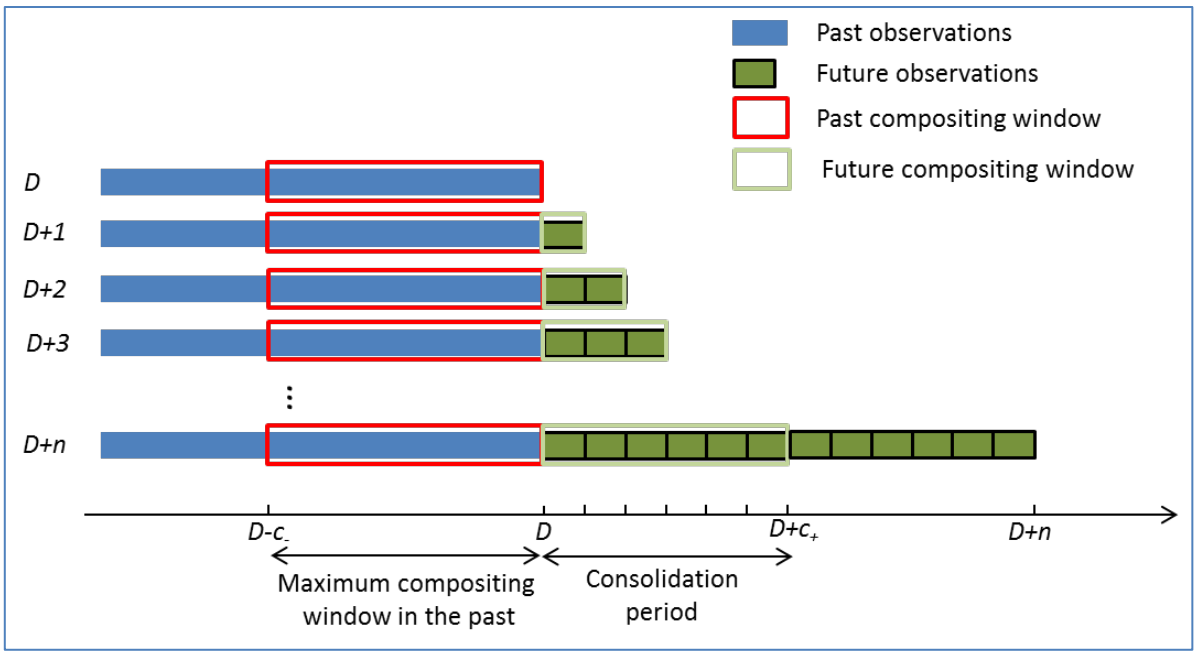


Figure 6: Scheme showing the compositing used for near real time estimates from RT0 (in the top) to the final consolidation RT6 (in the bottom). D refers to the dekad being processed. D+n corresponds to the n dekads available after the dekad D. c- refers to the maximum semi-compositing window before date D (60 days for nonEBF and 210 days for EBF) and c+ refers to the semi-compositing window after date D (0 days for RT0, 10 days for RT1, 20 days for RT2, 30 days for RT3 and 60 days for RT6 and after if the time series are reprocessed).

Note that when working in real time estimation and updating the values for each new dekad available, the compositing period is not symmetric: the maximum compositing window in the past is longer

before than after date *D*: $c_- > c_+$ for RT0, RT1, RT2…RT5. The compositing period after *D* is limited by the length of the consolidation period that is fixed to 60 days. For nonEBFs in RT6 case or when the time series are reprocessed in 'historical' mode after *D+6* dekads, the maximum length of the compositing period after date *D* should be equal to that in the period before, i.e. $c_+ = c_- = 60$ days (Figure 6). For EBFs, the compositing window is never symmetric: $c_-=210$ days and c_+=[0,60] days.

Once a new dekad is available after dekadal date D, the value of the RT product at dekadal date D is recomputed. Since the product value after the second consolidation (RT2) remains mostly stable (see section 5.1), the third, fourth and fifth consolidations are not distributed but only RT0, RT1, RT2 and the consolidated product after convergence, RT6, which is the final product. When the time series are reprocessed in 'historical' mode after *D+6* dekads, only RT6 products are produced and distributed. These are the main considerations for the processing of LAI, FAPAR, FCover 300m time series from PROBA-V (2014-2018) and Sentinel-3 (2019-present):

- Processing must be done from 2014 onwards, chronologically.
- Running PROBA-V chain from start of 2014 until end of 2019.
- Running Sentinel-3 chain from start of 2019 until present. Because of the compositing process, the processing should start only when 210 days have been accumulated. In practice, this means start running Sentinel-3 chain for 1st dekad 2019 using May-Dec 2018 Sentinel-3 data as input, i.e. generate daily LAI/FAPAR/FCover estimates from May 2018 (Step A of the algorithm) but generate composited values (Step B) only from start 2019 using the past May-Dec 2018 Sentinel-3 dailies for the composition.
- CCI-LC of the year 2014 is only used for the identification of EBFs at the beginning of the time series to initialize PROBA-V processing. Then the information for land cover estimation comes from the data. To initialize Sentinel-3 chain, EBF/No-EBF information from PROBA-V chain are used. This ensures the consistency in the transition from PROBA-V to Sentinel-3 at the start of 2019.

4 ALGORITHM DESCRIPTION

In this section, the inputs and outputs are described, along with the quality flags considered. Then, the several steps of the algorithm are presented in details.

4.1 INPUTS

All these inputs are required for each considered pixel.

4.1.1 Top of Canopy reflectance

The PROBA-V Collection 2 (C2) daily synthesis (S1) of Top of Canopy reflectance in B0: blue, B2: red and B3: near infrared bands (Table 1) are used as inputs. The SWIR band of PROBA-V sensor was not used because it is associated to a degraded spatial resolution and ground spatial sampling distance which are roughly twice that of the VIS-NIR domain (Figure 1). Note that the same spectral bands used in Version 1.0 of 300m products are used in Version 2.0.

The OLCI Top of Canopy reflectance in Oa4-12 and Oa16-18 spectral bands (Table 3) are used as inputs [CGLOPS1_ATBD_S3-AC-V1]. The blue bands Oa1-3 were discarded due to possible residual problems of atmospheric correction because they correspond to strong Rayleigh and aerosol scattering associated with low canopy reflectance levels (Liu et al. 2021). The narrow spectral bands Oa13-15 were discarded because they correspond to oxygen absorption bands used for aerosol, cloud and atmospheric correction. Finally, bands Oa19-21 were discarded because they are water absorption bands.

The PROBA-V and OLCI TOC reflectances should be expressed in terms of reflectance factor, mainly varying between 0.0 and 0.7 for most land surfaces outside the hot-spot or the specular directions and cloud, snow or ice cover.

4.1.2 Geometry of acquisition

The geometry information is required for the neural networks. It includes:

- the cosine of the view zenith angle (cos(VZA)),
- the cosine of the sun zenith angle (cos(SZA)),
- the cosine of the relative azimuth angle (cos(SAA-VAA)) where SAA corresponds to the Sun Azimuth Angle and VAA to the Viewing Azimuth Angle

4.1.3 Land cover

The CCI-LC (https://www.esa-landcover-cci.org) land cover map was used as auxiliary information for deciding whether a pixel belongs or not to an Evergreen Broadleaf Forest (EBF) at the beginning of the time series as described in Section 4.3.2.1. The CCI-LC (2.0.7cds) classification for the first

year of time series processing is used: i.e. CCI-LC 2014. This land cover map is derived from PROBA-V data at 300m (https://cds.climate.copernicus.eu/datasets/satellite-land-cover?tab=overview). According to CCI-LC legend, the EBF corresponds to class number 50 labeled as '*Tree cover, broadleaved, evergreen, closed to open (>15%)*'.

4.1.4 Land/water mask

A mask identifying both open ocean pixels and inland water pixels as "water" must be used as auxiliary information for deciding whether a pixel belongs to land or water. This mask is inherited from the quality flags of OLCI Sentinel-3 TOC reflectance. IDEPIX_LAND flag, from Idepix cloud detection module, and 'fresh_inland_water' flag, from OLCI L1B data, are used to distinguish "land" and "water" pixels. Only "land" pixels are processed. One flag (QC(1) in Table 8) associated to the LAI, FAPAR, FCover products specifies if the pixel is identified as land (QC(1)=0) or water (QC(1)=1).

4.1.5 Algorithm parameters

The two main steps (Step A and Step B) of the algorithm as showed in Figure 5 use a series of parameters listed in Table 4 and Table 5, respectively. Their roles and usages are further described in the various paragraphs of Section 4.3. The parameters used in the calibration of the neural networks and the elaboration of the definition domain are sensor specific and are given in Annex 1: Neural Network Calibration.

Table 4: The algorithmic parameters used in Step A.

Parameter	Descriptive Name	Type	Value	Reference section
$P^{tol}_{min_LAI}$	Tolerance limit minimum for LAI	Float	-0.2	4.3.1.4
$P^{tol}_{max_LAI}$	Tolerance limit maximum for LAI	Float	10	4.3.1.4
$P^{tol}_{min_FAPAR}$	Tolerance limit minimum for FAPAR	Float	-0.1	4.3.1.4
$P^{tol}_{max_FAPAR}$	Tolerance limit maximum for FAPAR	Float	1.2	4.3.1.4
$P^{tol}_{min_FCover}$	Tolerance limit minimum for FCover	Float	-0.1	4.3.1.4
$P^{tol}_{max_FCover}$	Tolerance limit maximum for FCover	Float	1.2	4.3.1.4

Table 5: The algorithmic parameters for Step B.

Parameter	Descriptive Name	Type	Units	Value	Reference section
$length_{EBF}^{\max_bef}$	Maximum length of the half compositing window before the date being processed for EBF case	Int.	days	210	4.3.2.1.1
$length_{EBF}^{\max_aft}$	Maximum length of the half compositing window after the date being processed for EBF case	Int	days	60	4.3.2.1.1
N_{EBF}	Number of observations required to compute products for EBF case	Int.	-	10	4.3.2.1.2
$f_x(N_{tot})$	Frequency used to consider an observation as valid for EBF case	Float	%	70	4.3.2.1.2
Lat_{EBF}^{max}	Latitude max at which EBF can be found (absolute value)	Float.	°	28.5	4.3.2.1.3
f_{EBF}^{min}	Ratio of number of dates at which the pixel is identified as EBF over the N_{EBF}^{max} dekads	Float	-	0.9	4.3.2.1.3
LAI_{EBF}^{min}	Threshold LAI value required to detect EBF (EBF should have LAI> LAI_{EBF}^{min})	Float	-	4	4.3.2.1.3
$Diff_{EBF}^{thres}$	Used to apply a threshold on the difference δ_{LAI_Valid} between consecutive valid observations to detect EBF	Float	-	0.9	4.3.2.1.3
$Percent_{EBF}$	Value of the percentile of difference between consecutive observations used to detect EBF	Float	%	80	4.3.2.1.3
$N_{outlier}^{min}$	Minimum number of observations required for outlier identification	Int	-	3	4.3.2.2.1
$tol_{outlier}^{abs}$	Value of outlier threshold (absolute value) used to detect outliers	Float	-	0.1	4.3.2.2.1
$tol_{outlier}^{rel}$	Value of outlier threshold (relative value) used to detect outliers	Float	-	0.6	4.3.2.2.1
$length_{no_EBF}^{outlier}$	Length of the half window used for outlier rejection for nonEBF case	Int.	days	5	4.3.2.2.1
$length_{no_EBF}^{min}$	Minimum length of the half compositing window for nonEBF case	Int.	days	15	4.3.2.2.2
$length_{no_EBF}^{max}$	Maximum length of the half compositing window for nonEBF case	Int.	days	60	4.3.2.2.2
N_{no_EBF}	Number of valid observations for nonEBF case in each half window used to define the length of the composition window	Int.	-	10	4.3.2.2.2
N_{linear}	Minimum number of valid observations for degree 2 polynomial fit, non EBF	Int.	-	5	4.3.2.2.3

k	Value of coefficient k in the weighing of observations for the polynomial fitting for non EBF case	Float	-	2	4.3.2.2.3
N_{miss}	Minimum number of observations for non_EBF case	Int.	-	3	4.3.2.2.3
ND_{gap}^{max}	Length in dekads of period with missing data that can be filled	Int.	dekads	6	4.3.2.2.3
$\Delta_{no_EBF}^{max}$	Maximum distance between the date of the dekad and the nearest observations required to compute the product in the non EBF case	Int.	days	15	4.3.2.2.3
$length_{no_EBF}^{interp}$	Length of the window used to interpolate between the 2 nearest data before and after dekad when nonEBF algorithm fails	Int.	days	15	4.3.2.2.3
$length_{no_EBF}^{nearest}$	Length of the window used to select the nearest data before or after dekad when nonEBF algorithm fails and interpolation is not possible	Int.	days	5	4.3.2.2.3
δ_{LAI}	Threshold value for the confidence interval on the estimated value	Float	-	0.35	4.3.2.2.3
tol_{FAPAR}^{min}	Tolerance minimum on FAPAR used to reject estimated values outside the expected range of variation	Float	-	-0.1	4.3.2.2.4
tol_{FCOVER}^{min}	Tolerance minimum on FCover used to reject estimated values outside the expected range of variation	Float	-	-0.1	4.3.2.2.4
tol_{LAI}^{min}	Tolerance minimum on LAI used to reject estimated values outside the expected range of variation	Float	-	-0.2	4.3.2.2.4
tol_{LAI}^{max}	Tolerance maximum on LAI used to reject estimated values outside the expected range of variation	Float	-	10	4.3.2.2.4
tol_{FAPAR}^{max}	Tolerance maximum on FAPAR used to reject estimated values outside the expected range of variation	Float	-	1.2	4.3.2.2.4
tol_{FCOVER}^{max}	Tolerance maximum on FCover used to reject estimated values outside the expected range of variation	Float	-	1.2	4.3.2.2.4
ND_{EBF}^{max}	Number of dekads used to compute the fraction of EBF cases	Int.	dekads	36	4.3.2.1.3

4.2 OUTPUTS

Three types of outputs are expected:

- The dekadal values of LAI, FAPAR and FCover

- Quantitative quality assessment (QA) indicators of the products
- Qualitative quality indicators (QC)

4.2.1 The LAI, FAPAR and FCover products

The outputs are computed by application of the algorithm over each pixel at each dekadal date. They include the LAI, FAPAR and FCover values as described previously. The range of variation and resolution are presented in Table 6. The same conventions as for 300m V1.0 products are used here. The definition of the maximum physical value of 1 for FCover is clear and corresponds to pixels with full cover green vegetation. The maximum FAPAR values are expected to be close to 0.94 (Baret and Guyot 1991) corresponding to full cover dense vegetation with albedo in the PAR domain close to 0.06. However, for LAI, the upper limit is not a physical limit, but a value just slightly higher than the maximum value that can be reached by the MODIS and CYCLOPES original products (Baret et al. 2013).

The physical values are retrieved by:

$$PhyVal = DN * Scaling_factor + Offset$$

where the scaling factor and the offset are given in Table 6.

Table 6: Minimum, maximum values and associated resolution for LAI, FAPAR and FCover products.

Product	Physical Minimum	Physical Maximum	Max DN value	Missing value	Scaling factor	Offset
LAI	0.0	7.0	210	255	1/30.	0
FAPAR	0.0	0.94	235	255	1/250.	0
FCover	0.0	1.0	250	255	1/250.	0

4.2.2 Quality indicators

In addition to the LAI, FAPAR and FCover values, quantitative (QA) and qualitative (QC) quality indicators are also generated. They are listed in Table 7 and Table 8, respectively.

The quantitative (QA) metrics NOBS, RMSE and LENGTH_BEFORE, LENGTH_AFTER are ancillary layers describing the quality of the product. The RMSE for each variable are calculated as the root mean square error between the instantaneous estimates in the compositing window and the final 10-day product value. See section 4.3.2 for more details.

Table 7: Minimum, maximum values and associated resolution for the quantitative quality indicators (QA) of LAI, FAPAR and FCover. *D* means dekadal date.

QA	Quantitative quality indicator	Physical Minimum	Physical Maximum	Max DN value	Scaling factor	Offset	Missing Value
NOBS	Number of available valid instantaneous estimates in the compositing window	0.0	60*	60	1	0	255
LENGTH_BEFORE	Length in days of semi-period before *D*	0	210	210	1	0	255
LENGTH_AFTER	Length in days of semi-period after *D*	0	60	60	1	0	255
RMSE-LAI	Uncertainty on LAI value	0.0	7.0	210	1/30.	0	255
RMSE-FAPAR	Uncertainty on FAPAR value	0.0	0.94	235	1/250.	0	255
RMSE-FCover	Uncertainty on FCover value	0.0	1.0	250	1/250.	0	255

** The theoretical maximum of NOBS is set to 60 to account for more than one instantaneous estimate per day in the compositing window but the typical maximum of NOBS is 10 for EBF and 20 for nonEBF.*

The qualitative quality flag (QC) indicator is coded as 8-bit (1 byte) pattern shown in Table 8. Bit 1 is the least significant bit (right-most). The QC value 255, i.e. 11111111 in binary form, is used for missing (non-processed) pixels.

Table 8: Qualitative quality indicators (QC). The numbers in brackets refer to the QC bit number. QC (1:8) = 11111111 indicates non-processed pixel.

QC	Qualitative quality indicator	Meaning	Reference section
1	Land/Water	QC (1) = 0: Land QC (1) = 1: Water	4.3.1.1
2	Vegetation class identified	QC (2) = 0: nonEBF case QC (2) = 1: EBF case	4.3.2.1.3
3	Not used	Not used	
4	Not used	Not used	
5	Method used for product computation EBF case	EBF case: QC (5) = 0: Based on instantaneous observations QC (5) = 1: Based on previous dekadal product or missing value (in this case the product value is set to missing value)	4.3.2.1.2
7:6	Method used for product computation nonEBF case	nonEBF case: QC (7:6) = 00: Second degree polynomials fit QC (7:6) = 10: Linear fit QC (7:6) = 01: Interpolation between the two nearest dates within $length_{no_EBF}^{interp}$ days QC (7:6) = 11: Nearest value within $length_{no_EBF}^{nearest}$ days or missing value (in this case the product value is set to missing value)	4.3.2.2.3
8	Instantaneous EBF classification flag	QC (8) = 0: nonEBF case instantaneous value QC (8) = 1: EBF case instantaneous value	4.3.2.1.3

4.3 DETAILED DESCRIPTION

As summarized in Figure 5 and detailed in Figure 7, the Version 2.0 algorithm of vegetation 300m products relies on 2 main steps:

- Step A: instantaneous estimates of LAI, FAPAR, FCover (Figure 8)
- Step B: compositing, smoothing and gap filling (gap filling is only applied for EBFs processing) (Figure 9)

The algorithm is dependent on the pixel land cover type (Figure 7). Specific Neural Network Techniques (NNTs) and dedicated smoothing procedures are applied, respectively, in steps A and B for Evergreen Broadleaf Forest (EBF) or non EBF pixels. The geographic position of the pixel is used first to trigger EBF or nonEBF branches. Only pixels located in tropical latitudes ($-28.5º \leq$

$Latitude \leq 28.5º$) or in Australia ($Latitude < 0º$ and $115º \leq Longitude \leq 155º$) are susceptible of being EBFs.

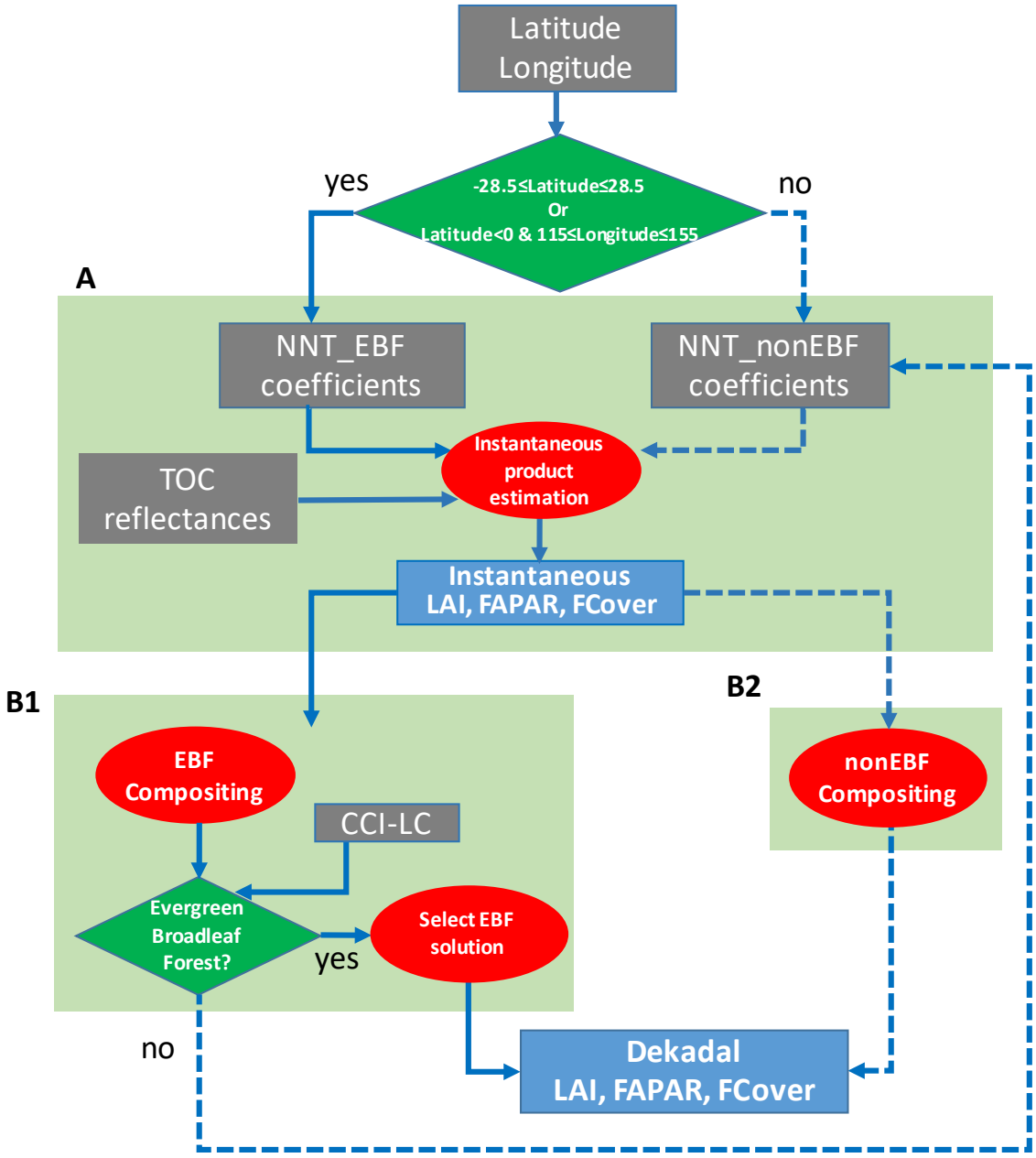


Figure 7: Detailed flow chart of 300m V2.0 algorithm. Solid line indicates the steps applied to EBF pixels. Dashed line corresponds to the steps applied to the nonEBF pixels.

For these potential EBF pixels, the algorithm is run in two iterations (Figure 7). At the first iteration, step A with NNT coefficients for EBFs and step B1 corresponding to EBF composition are applied. In addition to the dekadal estimates of LAI, FAPAR, FCover, step B1 also determines the land cover type of each pixel: Evergreen Broadleaf Forest (EBF) or nonEBF. The algorithm only keeps as EBFs those pixels in tropical latitudes or in Australia that, according to the data, have high LAI values, low seasonality and a significant level of noise. That is the algorithm itself searches and defines whether a pixel is EBF or nonEBF based on the LAI data values. However, at the beginning of the time series, the CCI-LC land cover is used as auxiliary information for initialization EBF and nonEBF masks. Note that this external land cover map is only used for processing the first year of the time series. The processing starts with historical PROBA-V time series followed by Sentinel-3 data, then the CCI-LC 2014 must be used. The information used for initialization EBF and nonEBF masks at the first dekad of Sentinel-3 time series is the EBF/nonEBF class resulted from Version 2.0 algorithm detection for the last dekad of PROBA-V processing. Further details of EBF/nonEBF detection are provided in section 4.3.2.1.3. If the pixel is detected as nonEBF, a second run is performed, with specific nonEBF NNTs in step A and step B2 smoothing procedure.

For the pixels which are not located in tropical latitudes or Australia, the algorithm is run in one single iteration using nonEBF branch (nonEBF NNTs in step A and step B2 for the composition) (Figure 7).

4.3.1 Instantaneous LAI, FAPAR, FCover estimates (Step A)

It yields a first estimate of instantaneous products from the Top-of-Canopy reflectance (Figure 8).

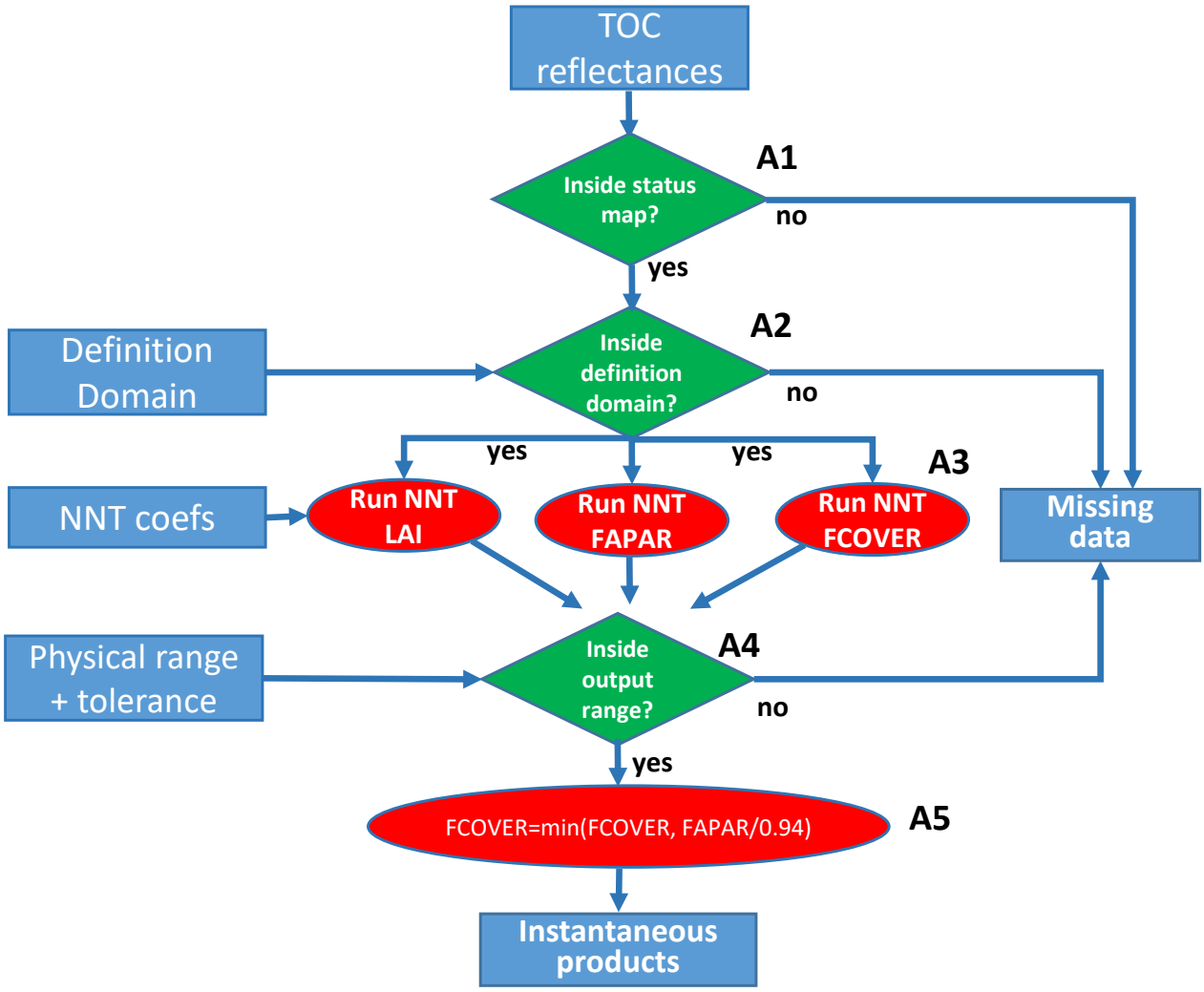


Figure 8: Flow chart describing the instantaneous product estimation (Step A)

4.3.1.1 Rejection of input data based upon their quality status (Step A1)

The quality flags associated to PROBA-V and Sentinel-3 TOC data are first used to keep only pixels with all the selected spectral bands having good radiometric quality, located over land, not covered by ice, cloud or snow.

For PROBA-V, the QC plane (Table 2) is first used to keep only the best quality pixels i.e. pixels with status map value equal to 248 = 11111000. Only land pixels, as identified by the land/water mask (Section 4.1.4), are processed.

For Sentinel-3:

- quality_flags layer inherited from OLCI L1B data (ACRI-ST, 2017; CGLOPS1_PUM_S3-TOC-Reflectance300m-V2.3):
  - 'saturated_Oa*' for bands Oa4-12 and Oa16-18, exclude when raised
  - For Sentinel-3 TOC reflectance V2.3 used in V2.0 algorithm, additional information is requested: 'fresh_inland_water', exclude when raised [CGLOPS1_PUM_S3-TOC-Reflectance300m-V2.3]. For Sentinel-3 TOC reflectances V1, used in V1.1 algorithm, this information was already included in the IDEPIX_LAND quality flag.
- pixel_classif_flags, output of Idepix cloud detection module, (CGLOPS1_QAR_S3-CloudMask, CGLOPS1_PUM_S3-TOC-Reflectance300m-V2.3):
  - IDEPIX_LAND: include when raised
  - IDEPIX_CLOUD: exclude when raised
  - IDEPIX_CLOUD_AMBIGUOUS: exclude when raised
  - IDEPIX_CLOUD_BUFFER: exclude when raised
  - IDEPIX_CLOUD_SHADOW: exclude when raised
  - IDEPIX_SNOW_ICE: exclude when raised

4.3.1.2 First outlier rejection (Step A2)

A first condition is applied to verify whether the inputs of a given observation keep within the range of variation of the training dataset called here the definition domain. In order to avoid that the Neural networks would extrapolate values, only the input TOC reflectance in the selected spectral bands for a given observation which are within the definition domain (Table 9, Table 10, Annex 1: Neural Network Calibration) are kept as valid input data. The observations with TOC data values out of the definition domain are considered as outliers and rejected. Note that the definition domain is sensor specific: one domain is defined for PROBA-V and one domain for Sentinel-3.

4.3.1.3 Deriving instantaneous estimates using neural networks (Step A3)

Two sets of specific neural networks for EBF and nonEBF were previously calibrated for each of the 3 variables considered (LAI, FAPAR, and FCover) (Annex 1: Neural Network Calibration). The neural networks are also sensor specific: one set of neural networks is calibrated for PROBA-V and other for Sentinel-3. Then the combination of three variables (LAI, FAPAR, FCover) per two landcover vegetation classes (EBF and nonEBF) per two sensors (PROBA-V and Sentinel-3) resulted in twelve neural networks.

For the pixels located outside the regions of the world suitable for EBF (Figure 7), nonEBF NNTs are applied to each individual observation (one pixel at a given date).

For the pixels located in a region of the world suitable for EBF, the EBF NNTs are used similarly. If the EBF identification achieved in Step B1 (Figure 7) reveals that some of these pixels are actually not EBF then, the nonEBF NNTs are used to reprocess these pixels in a second iteration (Figure 7).

The inputs of the neural networks are:

- TOC reflectance in the selected bands (the PROBA-V neural networks ingest TOC reflectance in 3 spectral bands: B0, B2 and B3, while the Sentinel-3 neural networks ingest TOC reflectance in 12 OLCI spectral bands: Oa4-12 and Oa16-18),
- the cosine of the view zenith angle (cos(VZA)),
- the cosine of the sun zenith angle (cos(SZA)),
- the cosine of the relative azimuth angle (cos(SAA-VAA)),

To apply the neural networks, the following steps must be completed:

- **Normalization of the inputs**: The inputs are normalized to prevent possible numerical problems during the training process. For all the inputs X, the following normalization equation must be applied:

$$Xnorm = 2.(X - Xmin)/(Xmax - Xmin) \; - \; 1$$

  where $Xnorm$ is the normalized input value, and $Xmin$ and $Xmax$ correspond to the minimum and maximum values of the inputs in the neural network training data set (Table 9, Table 10).

- **Run the neural network.** The neural network is described by its architecture, i.e. the number of hidden layers and the output layer. Each layer is described by its number of neurons, associated weight and biases and transfer function. A simple neural network with one hidden layer with 5 neurons and one output layer was used. For the neurons of the hidden layers, the transfer function is a tangent sigmoid function given by: $y = Tansig(x) = 2/(1 + exp(-2x)) - 1$, while for the output layer the transfer function is linear ($y = x$) .

- **Denormalization of the output.** It simply consists in applying the inverse function used for input normalization:

$$Y = 0.5.(Ynorm + 1) * (Ymax^* - Ymin^*) \; + Ymin^*$$

where $\boldsymbol{Ynorm}$ is the normalized output value issued from the NNT, and $\boldsymbol{Ymin}^*$ and $\boldsymbol{Ymax}^*$ are computed over the neural network training data set (Table 9, Table 10).

Table 9: Minimum and maximum values used for the normalization of inputs and outputs of nonEBF and EBF neural networks for PROBA-V.

		nonEBF		EBF	
		Min	Max	Min	Max
Inputs	B0	0.00183	0.53400	0.00050	0.73317
	B2	0.01250	0.70843	0.00817	0.80617
	B3	0.05100	0.81736	0.05100	0.83883
	cos(VZA)	0.48481	0.99966	0.48481	0.99966
	cos(SZA)	0.25882	0.98325	0.17365	0.98325
	cos(SAA-VAA)	-1.00000	1.00000	-1.00000	1.00000
Outputs	LAI	0	6.69993	0	6.95326
	FAPAR	0	0.95628	0	0.98019
	FCover	0	1.01732	0	1.04276

Table 10: Minimum and maximum values used for the normalization of inputs and outputs of nonEBF and EBF neural networks for Sentinel-3.

		nonEBF		EBF	
		Min	Max	Min	Max
Inputs	Oa04	0.0001	0.6335	0.0002	0.6335
	Oa05	0.0013	0.6389	0.0039	0.6389
	Oa06	0.0086	0.7602	0.0129	0.7602
	Oa07	0.0059	0.8553	0.0098	0.8553
	Oa08	0.0052	0.8763	0.0078	0.8763
	Oa09	0.0048	0.8870	0.0073	0.8870
	Oa10	0.0049	0.8874	0.0075	0.8874
	Oa11	0.0165	0.8903	0.0215	0.8903
	Oa12	0.0357	0.9475	0.0357	0.9475
	Oa16	0.0355	0.9515	0.0355	0.9515
	Oa17	0.0338	0.9627	0.0338	0.9627
	Oa18	0.0433	0.9893	0.0433	0.9893
	cos(VZA)	0.5694	0.9999	0.5694	0.9999
	cos(SZA)	0.2187	0.9366	0.1709	0.9380
	cos(SAA-VAA)	-1.0000	1.0000	-1.0000	1.0000
Outputs	LAI	0	6.8466	0	6.8666
	FAPAR	0	0.94	0	0.94
	FCover	0	1	0	1

4.3.1.4 Output outlier rejection (Step A4)

The instantaneous values lying outside the physical range of variation of the variables extended by the tolerance limits (Table 11) are rejected. Three cases of the variable values are possible:

1. The value is within expected physical range of variation (Table 6): it is considered valid.
2. The value is within the tolerance limits (Table 11) but higher (lower) than the physical maximum (minimum) (Table 6). The value is considered valid but set to the physical maximum (minimum).
3. The value is outside the tolerance limits: it is considered invalid.

If one of the variables is considered invalid, all the three variables are rejected. This allows keeping a high level of consistency between the three products (LAI, FAPAR and FCover).

Table 11: Physical range of variation and tolerance limits (Minimum (P^{tol}_{min}) and maximum (P^{tol}_{max})) used to reject the output.

	LAI	FAPAR	FCover
P^{tol}_{min}	-0.20	-0.1	-0.1
P^{tol}_{max}	10.0	1.2	1.2
Physical Minimum	0	0	0
Physical Maximum	7	0.94	1

4.3.1.5 Ensuring FAPAR-FCover consistency (Step A5)

The FAPAR value divided by its maximum theoretical value of 0.94 (Baret and Guyot 1991) must remain higher than FCover in any situations since it can be approximated by the FIPAR (Fraction of Intercepted Photosynthetically Active Radiation). Indeed, FIPAR is the complementary of the gap fraction at a higher angle than nadir (that corresponds to FCover). Therefore, to avoid possible physical inconsistencies between the estimated FAPAR and FCover values, a final condition must be set to the $FAPAR_{NNET}$ and $FCover_{NNET}$ outputs of neural networks:

$$FCover = \min\left(FCover_{NNET}, \frac{FAPAR_{NNET}}{0.94}\right)$$

This way, a corrected FCover is generated ensuring the consistency with FAPAR.

The resulting time series after Step 5A are the instantaneous LAI, FAPAR and FCover estimates (Figure 8).

4.3.2 Compositing, smoothing and gap filling (Step B)

The compositing, filtering and gap filling module is run for each dekadal date *D*. Two cases are considered: B1 for Evergreen Broadleaf Forest (EBF), and B2 for non-Evergreen Broadleaf Forest (nonEBF). The gap filling is only applied for EBF processing.

Note that similarly to Step A, the algorithm produces in parallel the three variables (LAI, FAPAR and FCover) since the final dekadal products are strongly inter-related.

4.3.2.1 Run Evergreen Broadleaf Forest case (Step B1)

Evergreen Broadleaf Forests are characterized by:

- A geographic location within the tropical area
- A low seasonality
- A large amount of cloud contaminated observations.

The processing of such situations needs to take into account explicitly these peculiarities. The algorithm includes four steps for each dekadal date:

1. Adjusting the length of the compositing window (Step B1.1)
2. Computing the variables estimate for the dekadal date (Step B1.2)
3. Deciding whether the pixel belongs actually to the EBF or nonEBF class (Step B1.3)
4. Computing the uncertainties associated to the estimates (Step B1.4)

4.3.2.1.1 Adjusting the length of the compositing window (Step B1.1)

First, a large compositing window is considered. It is defined as the maximum length (in days) of the period before the dekadal date ($length_{EBF}^{\max_bef}$), and the maximum length of the period after this date ($length_{EBF}^{\max_aft}$). $length_{EBF}^{\max_aft}$ is lower than $length_{EBF}^{\max_bef}$ to prevent from having a too long consolidation period while being consistent between historical processing and near real time situations. We therefore chose the same value as for the nonEBF case (section 4.3.2.2). $length_{EBF}^{\max_bef}$ is chosen in order to avoid missing values for the first dates of the time series. Due to high cloud occurrence and missing data, a long period is required for processing EBF pixels for which we assume low seasonality. As default values, we use $length_{EBF}^{\max_bef} = 210\ days$ and $length_{EBF}^{\max_aft} = 60\ days$. For near real time processing, $length_{EBF}^{\max_aft} = \min(t -$

$D, length_{EBF}^{\max_aft}$) where t corresponds to the date when the latest instantaneous product value available, and D corresponds to the dekadal date being processed.

The N_{tot} observations the closest to dekadal date D are selected within the available 'valid' instantaneous estimates in the compositing window. N_{tot} corresponds to the total number of selected observations: $N_{tot} = N_{before} + N_{after}$ where N_{before} and N_{after} refer to the selected observations in the semiperiod before and after the date D of the product, respectively. $N_{tot} \leq N_{EBF}$ with $N_{EBF} = 10$, i.e. we select up to 10 observations the closest to the decadal date D. For EBFs, N_{tot} corresponds to the NOBS metrics provided in the QA indicators (Table 7). Similarly, after the section of instantaneous estimates, the length of semi-period of composition are fixed and reported as the LENGTH_BEFORE and LENGTH_AFTER QA layers (Table 7). LENGTH_BEFORE is the distance in days between the date of the first observation in the compositing window and the decadal date (D) of the composited product. LENGTH_AFTER is the distance in days between the date of the last observation in the compositing window and the dekadal date (D) of the composited product.

4.3.2.1.2 Computing the dekadal estimates (Step B1.2)

Two cases are considered depending on the available number of observations, $N_{tot} = N_{before} + N_{after}$in the compositing window (Figure 9):

1. ***If $N_{tot} = N_{EBF}$, computing the mean of 'valid' instantaneous estimates.*** A frequential filter is first applied to discard the low values corresponding to cloud contamination: date t of an instantaneous observation is considered valid if $LAI(t) \geq f_x(N_{tot})$, where $f_x(N_{tot})$ is the percentile x value computed on the available total number of instantaneous observations over the compositing window, N_{tot}. x was set to $x = 70\%$. Then, the mean computed over the valid observations and assigned to the dekadal date D:

$$If\, N_{tot} = N_{EBF}, P(D) = mean(P(t), with\, LAI(t) \geq f_x(N_{tot}))$$

2. ***If $N_{tot} < N_{EBF}$*** within the compositing window, the product value of dekadal date D is set to the previous EBF product value:

$$If\, N_{tot} < N_{EBF}, P(D) = P(D-1)$$

   For RT processing, we consider the previous dekadal value estimated in the consolidated mode being processed.

   If there was missing data for the previous dekad, then a missing value is set to the dekadal date D.

Note that the proposed approach propagates in time the last valid EBF product value computed in case 1 using $N_{tot} = N_{EBF}$ to the next dekads for which $N_{tot} < N_{EBF}$ (case 2). A quality indicator (QC(5) in Table 8) specifies if the product value is actually computed based on the instantaneous

observations considered for dekadal date *D* (QC(5)=0, case 1), or if it is estimated using the previous dekadal value or missing value (QC(5)=1, case 2).

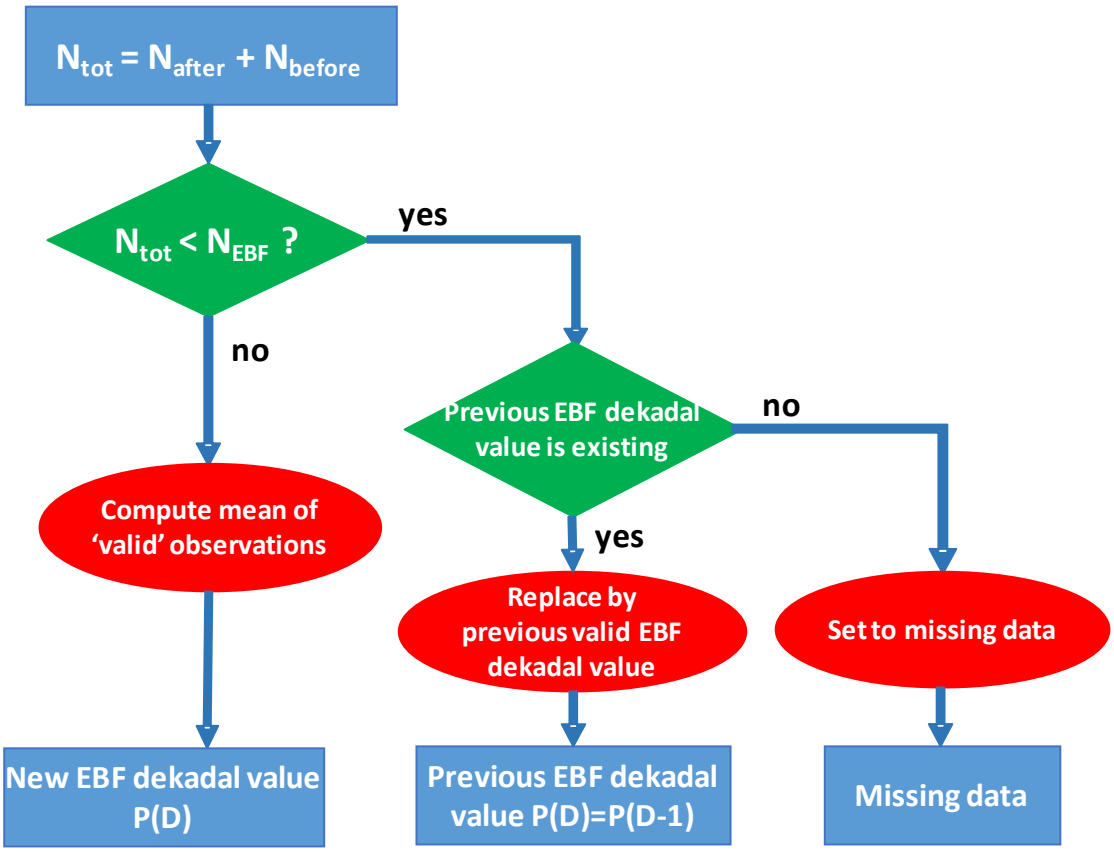


Figure 9: Flow chart showing the different cases for smoothing, and gap filling at the dekadal date D for Evergreen Broadleaf Forest situations. D indicates the current dekad at which the product is computed (D-1 being the previous dekad).

4.3.2.1.3 Checking the land cover (Step B1.3)

The use of an ancillary classification, such as CCI-LC, was a possible solution. However, it poses at least three problems:

1. The grid used for CCI-LC (1/360°) is different from that used for Sentinel-3 TOC data.
2. The classification map is fixed on an annual basis, although some pixels may change due to deforestation or reforestation,
3. The classification map may incorporate some uncertainties

For this reason, we preferred using the PROBA-V and Sentinel-3 observations directly to identify the EBF cases when possible. However, at the beginning of the time series for the first year of PROBA-V, when the data are either too few or too scattered, the CCI-LC land cover is used as auxiliary information (Section 4.1.3).

We first estimate an 'instantaneous' land cover class $LC_D(D)$ that is then regularized using the LC_D values estimated in the previous dekads within the period $D - ND_{EBF}^{max} : D$ with $ND_{EBF}^{max} = 36\ dekads$ (Figure 10, Figure 11). This results in the final land cover estimate $LandCover_{Est}(D)$ for a given pixel and dekad D (Figure 11).

I. Instantaneous land cover $LC_D(D)$

We estimate the instantaneous land cover class $LC_D(D)$ of a given pixel as an EBF using two criteria that must be simultaneously fulfilled:

1. The estimated LAI value should be higher than LAI_{EBF}^{min} with default value $LAI_{EBF}^{min} = 4.0$ since EBF are characterized by high LAI values (Figure 16, top).
2. There is a significant level of noise in the instantaneous LAI temporal profile. The temporal smoothness of time series of instantaneous LAI estimates is used as a metrics to evaluate the level of noise. The percentile $Percent_{EBF}$ (with default value $Percent_{EBF} = 80\%$) of the distribution of the absolute value of the difference between two consecutive valid LAI observations (δ_{LAI_Valid}) is computed over the temporal window used for EBF case as defined in Step B1.1. Indeed, high differences between two close dates correspond to situations with high probability of cloud contamination and induce high variation in the data. Then, if the percentile $Percent_{EBF}$ of δ_{LAI_Valid} is higher than the threshold value $Diff_{EBF}^{thres}$, $Diff_{EBF}^{thres} = 0.9$ being the default value, and condition 1 is also fulfilled, the pixel is classified as an EBF.

Different particular cases are considered depending on the location of the pixel, the number of available dekads and the number of valid instantaneous estimates in the compositing window (Figure 10):

- The pixels with latitudes $Lat > Lat_{EBF}^{max}$ (where $Lat_{EBF}^{max} = 28.5°$), or $Lat < -Lat_{EBF}^{max}$ and which are not in Australia ($Latitude < 0º$ and $115º \leq Longitude \leq 155º$) are considered as being nonEBFs.
- At the beginning of the time series, when the number of available dekads $\boldsymbol{ND_{available} < ND_{EBF}^{max}}$ with $ND_{EBF}^{max} = 36\ dekads$, the CCI-LC classification is used to fill the instantaneous land cover $LC_D(D)$ for the first $ND_{EBF}^{max} - ND_{available}$ missing dekads (Figure 11).

- When there are not enough valid instantaneous estimates in the compositing period, $\boldsymbol{N_{tot}} < N_{EBF}$, the previous instantaneous land cover estimate is used to fill the landcover at the dekad D ($LC_D(D) = LC_D(D-1)$) (Figure 10).

One flag (QC(8) in Table 8) corresponds to instantaneous land cover, $LC_D(D)$: QC(8)=1 if the pixel at dekad D is detected as EBF.

II. Regularized land cover *LandCover$_{Est}$(D)*

Because of the generally high cloud cover and contamination, the 'instantaneous' estimates of land cover derived from the 3 previous criterions are regularized using the estimates from the previous dekads (Figure 11). As a matter of fact, the land cover class of a pixel is not likely to change often with time. The instantaneous land cover *LC* estimates of previous ND_{EBF}^{max} dekads are used. For RT processing, we consider the previous land cover class computed in the specific consolidated mode being processed. The frequency of dekads assigned to EBF in the previous period of $(D - ND_{EBF}^{max} : D)$ dekads is computed (f_{EBF}). The default value of ND_{EBF}^{max} was set to $ND_{EBF}^{max} = 36\ dekads$ (≈360 days) and f_{EBF}^{min} was set to $f_{EBF}^{min} = 0.9$.

- If the frequency is:
  - $f_{EBF} \leq 1 - f_{EBF}^{min}$ then the *LandCover$_{Est}$(D)* is considered as non_EBF
  - $f_{EBF} \geq f_{EBF}^{min}$ then the *LandCover$_{Est}$(D)* is considered as EBF.
- If this frequency is inside the range $1 - f_{EBF}^{min} < f_{EBF} < f_{EBF}^{min}$, the estimates from the observations are considered unreliable and the previous land cover estimation is used: *LandCover$_{Est}$(D)= LandCover$_{Est}$(D-1)*. For RT processing, we consider the *LandCover$_{Est}$(D-1)* computed in the consolidated mode being processed.

At the beginning of the time series at dekad D, when the number of available dekads $\boldsymbol{ND_{available}} < \boldsymbol{ND_{EBF}^{max}}$ with $ND_{EBF}^{max} = 36\ dekads$, the CCI-LC classification is used to identify *LandCover$_{Est}$(D)* and to complement LD estimates for the available dekads, *i.e. LandCover$_{Est}$(D-ND_{EBF}^{max}: D-$ND_{available}$-1)=CCILC* and *LD(D-ND_{EBF}^{max}: D-$ND_{available}$-1)=CCILC* (Figure 11). Note the CCI-LC is only used for the first year of the time series of the input data. The global annual CCI-LC (2.0.7cds) map for the year 2014 is used. CCI-LC 2014 corresponds to the first year of PROBA-V processing. Then the information to estimate the land cover comes from the data. For the initialization of Sentinel-3 processing at the beginning of the time series in the year 2019, when $ND_{available\ from\ Sentinel-3} < ND_{EBF}^{max}$ with $ND_{EBF}^{max} = 36\ dekads$, instantaneous LC and regularized *LandCover$_{Est}$* for the dekads *D-ND_{EBF}^{max}: D-$ND_{available}$-1* result from PROBA-V chain. This ensures a continuity between PROBA-V and Sentinel-3 time series.

One flag (QC(2) in Table 8) is then set to report the land cover ($LandCover_{est}(D)$) that is finally assigned, when considering the f_{EBF} value computed over the previous ($D - ND_{EBF}^{max}: D$) dekads (QC(2)=1 for EBF).

After regularization of the EBF land cover, if the bit 2 of QC is activated (QC(2)=1) the outputs of the EBF algorithm are kept as the final product for date *D*. Otherwise, if QC(2)=0, i.e the land cover of that particular pixel at date *D* is detected as nonEBF, a second run is performed using the nonEBF algorithm.

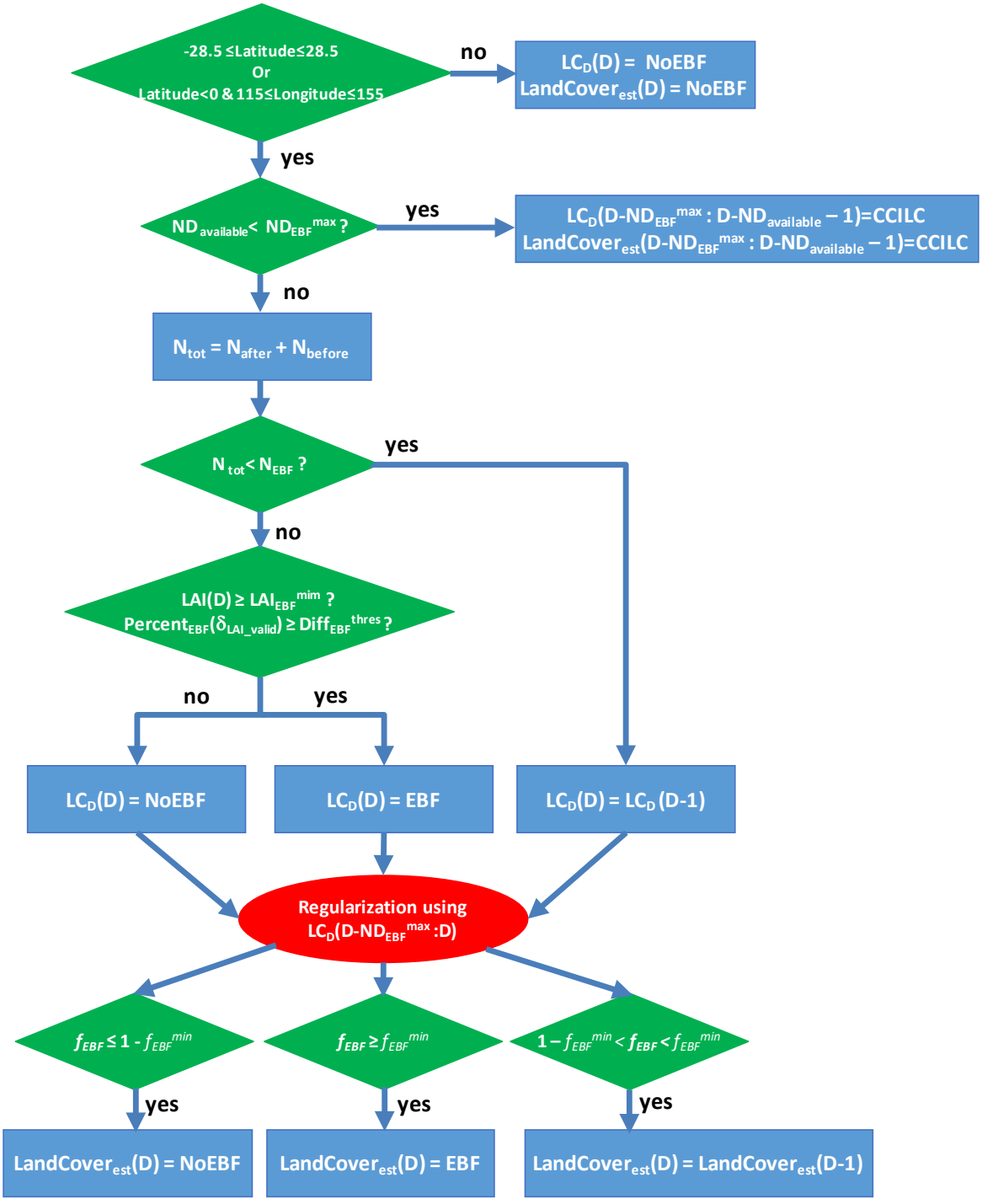


Figure 10: Flow chart showing the land cover class identification after applying the EBF compositing algorithm. At the beginning of the time series, the land cover is set to the CCI-LC class (EBF or nonEBF). The computation of f_{EBF} is illustrated in Figure 11. D indicates the current dekad at which the product is computed (D-1 being the previous dekad).

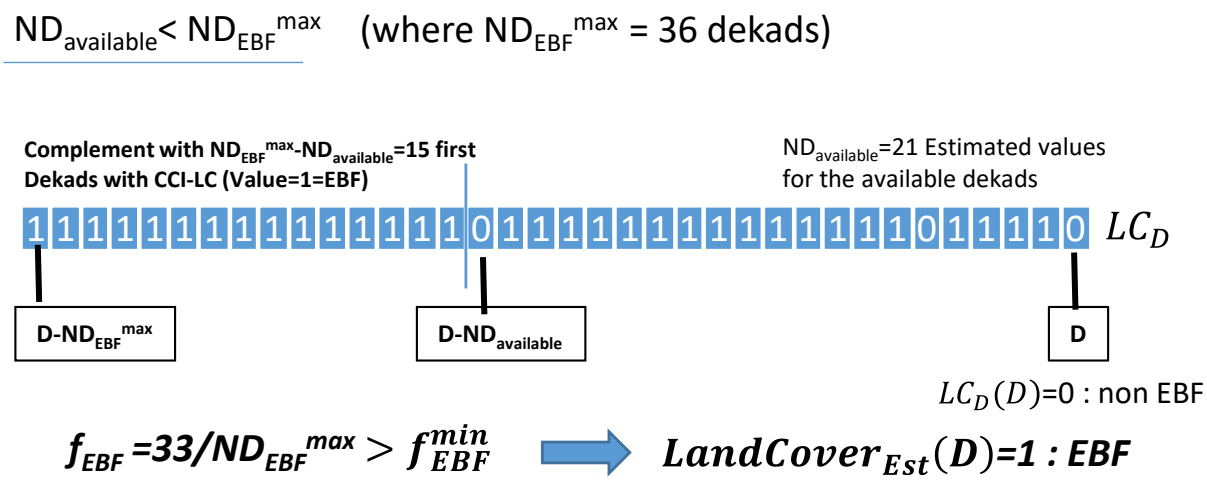


Figure 11: Example of land cover class determination criterion: only the 21st dekads are available. The instantaneous land cover at dekadal date D is non EBF ($LD_D(D)=0$). After regularization using the estimates from the previous $(D - ND_{EBF}^{max} : D)$ dekads, the computed frequency of dekads assigned to EBF $f_{EBF} > f_{EBF}^{min}$ and then the final class attributed to this pixel at dekad D is EBF ($LandCover_{Est}(D)=1$).

4.3.2.1.4 Computing the uncertainties for EBF (Step B1.4)

Two cases are considered depending on the number of available instantaneous valid estimates in the compositing window:

- If $N_{tot} = N_{EBF}$, the uncertainties are computed as the Root Mean Square Error (RMSE) between the instantaneous valid estimates of the product $P(t)$ after filtering and the product value at dekad D, $P(D)$, as described in Step B1.2:

$$RMSE = \sqrt{\frac{\sum_{t=1}^{n_{valid}} (P(t) - P(D))^2}{n_{valid}}}$$

  Where n_{valid} is the number of valid instantaneous estimates $P(t)$ used to compute $P(D)$. A valid estimate $P(t)$ corresponds to the instantaneous values after the LAI filtering described in Step B1.2 (instantaneous $P(t)$ values at time t are valid if $LAI(t) \geq f_x(N_{tot})$).

- If $N_{tot} < N_{EBF}$, the uncertainty at D is set to missing value, i.e. 255.

4.3.2.2 Non-Evergreen Broadleaf Forest case (Step B2)

This step of the algorithm is triggered for non EBFs. This may correspond to two cases (Figure 7) (i) pixels not located at $-28.5º \leq Latitude \leq 28.5º$ nor in Australia ($Latitude < 0º$ and $115º \leq Longitude \leq 155º$), or (ii) after a first run as EBF, the estimated land cover at dekad D (output of Step B1.3), $LandCover_{est}(D)$, is finally non EBF.

Five steps are considered:

1. First, outliers are rejected based on the smoothness of the temporal profiles of instantaneous estimates (Step B2.1).
2. Second, the compositing window is adjusted (Step B2.2).
3. Then, the valid instantaneous observations within the compositing window are used to fit a linear or second-degree polynomial depending on the number of available instantaneous observations (Step B2.3).
4. A post-filtering is applied to reject outputs outside the variable range of physical values including tolerance (Step B2.4).
5. Finally, the uncertainties are calculated (Step B2.5).

4.3.2.2.1 Rejecting outliers based on expected smoothness of instantaneous estimates time series (Step B2.1)

During the outlier rejection, emphasis was put on LAI products that show the highest sensitivity to possible problems in reflectance values. Therefore, when an outlier is detected on LAI data, it is also considered as an outlier for FAPAR and FCover to keep a high level of consistency between the three variables. Note that possible outliers in FAPAR or FCover but not LAI will not be filtered. This criterion allows reducing computation time which is key to run operationally this relatively complex algorithm in near real time.

The consistency of each instantaneous estimate with regards to the before and after observations is evaluated using a temporal window of half-length equal to $length_{no_EBF}^{outlier}$ and centered on the considered instantaneous estimate. The default value for $length_{no_EBF}^{outlier}$ is set to $length_{no_EBF}^{outlier} = 5\ days$. However, if there are less than $N_{outlier}^{min}$ instantaneous estimates within the temporal window, no outlier rejection is considered. The default value of $N_{outlier}^{min}$ is set to 3. The maximum LAI values and associated dates observed in the $length_{no_EBF}^{outlier}$ half-length temporal window before (LAI_{before}^{max}) and after (LAI_{after}^{max}) are identified. The maximum LAI value is used here since it is the least affected by possible cloud contamination. Then, the value of the considered observation, LAI, is compared to the value interpolated, LAI_{interp}, at the same date between LAI_{before}^{max} and after LAI_{after}^{max} to detect positive (negative) spikes to be filtered:

- Positive spike: $LAI \geq LAI_{interp} + \max\left(tol_{outlier}^{abs}, tol_{outlier}^{rel}.LAI_{interp}\right)$
- Negative spike: $LAI \leq LAI_{interp} - \max\left(tol_{outlier}^{abs}, tol_{outlier}^{rel}.LAI_{interp}\right)$

where $tol_{outlier}^{abs}, tol_{outlier}^{rel}$ are respectively the absolute and relative values of the tolerance, with $\left[tol_{outlier}^{abs}, tol_{outlier}^{rel}\right] = [0.1, 0.6]$ as default values. If a spike is detected for the considered observation, the instantaneous estimates (LAI, FAPAR, FCover) are rejected resulting in missing data for this date of observation.

4.3.2.2.2 Adjusting the length of the compositing window (Step B2.2)

An extended compositing period is first defined to filter data and select the closest data to the dekad D within the available instantaneous estimates. The compositing window for a given dekadal date D is made of two parts: the sub-window before date D and the sub-window after date D. The maximum length of each sub-window is set to $length_{no_EBF}^{max} + length_{no_EBF}^{outlier}$ (in days). The value of $length_{no_EBF}^{max}$ was tuned and fixed to $length_{no_EBF}^{max} = 60\ days$. The additional period $length_{no_EBF}^{outlier}$ accounts for the window required for local outlier rejection.

After outlier rejection, the actual length of each sub-window and the corresponding number of available instantaneous estimates used for the composition are defined by:

- ***The sub*-window before dekadal date *D*.** The length of the sub-window before dekadal date D, $length_{before}$ (in days), is defined by the period that contains the N_{no_EBF} observations which are the closest to dekadal date D. The value of N_{no_EBF} is set to $N_{no_EBF} = 10$ as default. When the number of observations in the vicinity of the date D is higher than N_{no_EBF}, *i.e.* $N_{before} > N_{no_EBF}$, a minimum length of the sub-window, $length_{no_EBF}^{min}$, is considered to increase the robustness of the fitting. The value of $length_{no_EBF}^{min}$ is set to $length_{no_EBF}^{min} = 15\ days$. When no data is available before dekadal date D within the maximum $length_{no_EBF}^{max}$ semi-period of composition, $N_{before} = 0$ and $length_{before} = 0$. The value of $length_{no_EBF}^{max}$ is set to $length_{no_EBF}^{max} = 60\ days$. The actual number of final observations N_{before} and $length_{before}$, $0 \leq length_{before} \leq length_{no_EBF}^{max}$, depends on the available observations in the vicinity of the date D for the cases described in Figure 12.
- ***The sub*-window after dekadal date *D*.** Because we are considering here near-real time products, the length of the sub-window after dekadal date D, $length_{after}$ (in days), will depend on the considered date D within the consolidation period and the last date at which an instantaneous product is available. $length_{after}$ is defined as the period within $length_{no_EBF}^{max}$ which contains the closest N_{no_EBF} instantaneous estimates to dekadal date D. The value of N_{no_EBF} is the same as for the temporal window used before the dekadal date D. A minimum length of the sub-window, $length_{no_EBF}^{min}$, is also considered to increase the

robustness of the fitting when $N_{after} > N_{no_EBF}$ in the vicinity of D. The value of $length_{no_EBF}^{min}$ is set to the same value as the one used for the temporal window before dekadal date D, i. e. $length_{no_EBF}^{min} = 15\ days$.

Obviously, when no data is available after dekadal date D, including the processing the RT0 products, $length_{after} = 0\ days$. When less than two dekads are available after date D, the actual value of $length_{after}$ is set to $length_{after} = t - D$, where t corresponds to the latest instantaneous product value available. Otherwise, when two or more dekads are available after date D for near real time mode, $length_{after}$ corresponds to the period within $length_{no_{EBF}}^{min} \leq length_{after} \leq t - D$ which contains the closest N_{no_EBF} instantaneous estimates to dekadal date D. The actual length of the sub-window after is therefore fluctuating, with $0 \leq length_{after} \leq \min(t - D, length_{no_{EBF}}^{max})$, depending on the available instantaneous estimates in the vicinity of the considered dekadal date (Figure 13).

Then, $length_{before}$ corresponds to the distance in days between the date D of the current dekad and the date of the first observation in the compositing window, and $length_{after}$ to the distance between the date of the last observation and D. They are respectively reported as the LENGTH_BEFORE and LENGTH_AFTER metrics in the QA indicators for nonEBFs (Table 7). The available number of observations, $N_{tot} = N_{before} + N_{after}$, is reported as NOBS metrics (Table 7).

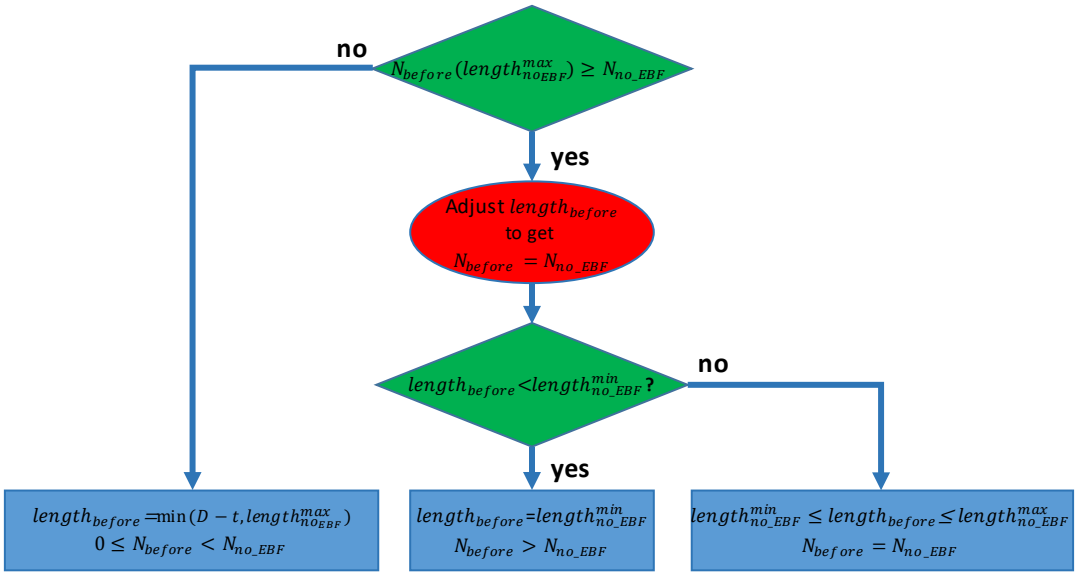


Figure 12: Scheme describing how to set the values $length_{before}$ and N_{before}. D indicates the date of the current dekad at which the product is computed, and t *is* the first date within D-$length_{no_EBF}^{max}$ at which an instantaneous product is available.

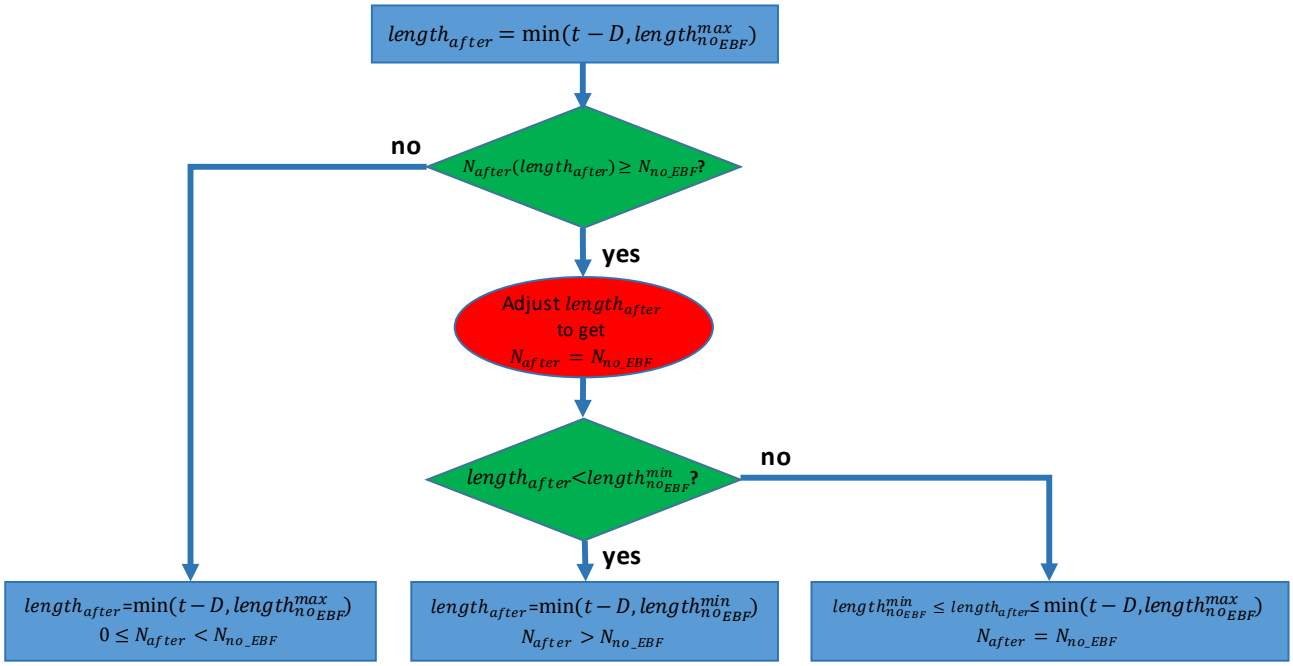


Figure 13: Scheme describing how to set the values $length_{after}$ and N_{after}. D indicates the current dekad at which the product is computed, and t is the latest date at which an instantaneous product is available.

This general rule applies to all the 4 cases encountered, i. e. for:

1. ***The start of the time series***, when no data is available before dekadal date D. In this case the processing should start only when $length_{no_EBF}^{max}$ days have been accumulated.
2. ***Near real time RT0 estimates***, when no data is available after dekadal date D
3. ***The near real time RT1-RT6 in the consolidation period***, when only few observations are available after dekadal date D. The consolidation period lasts $length_{no_EBF}^{max}$ days.
4. ***Historical processing***, when dekadal date D is followed by observations available for the next $length_{no_EBF}^{max}$ days.

4.3.2.2.3 Computing the dekadal estimates (Step B2.3)

The temporal compositing, smoothing and projection is achieved based on a low degree polynomial fit. The degree of the polynomials applied over the compositing window is modulated depending on the available number of observations, $N_{tot} = N_{before} + N_{after}$.

A first test is made to detect if the nearest observations within the compositing window is sufficiently close to the dekad date (dekad-nearest_observation $\leq \Delta_{no_EBF}^{max}$). $\Delta_{no_EBF}^{max}$ is set to 15. If this is not the case, no value is computed for this dekad and the product values for LAI, FAPAR and FCover are set to missing data (=255); otherwise, three cases are considered (Figure 14):

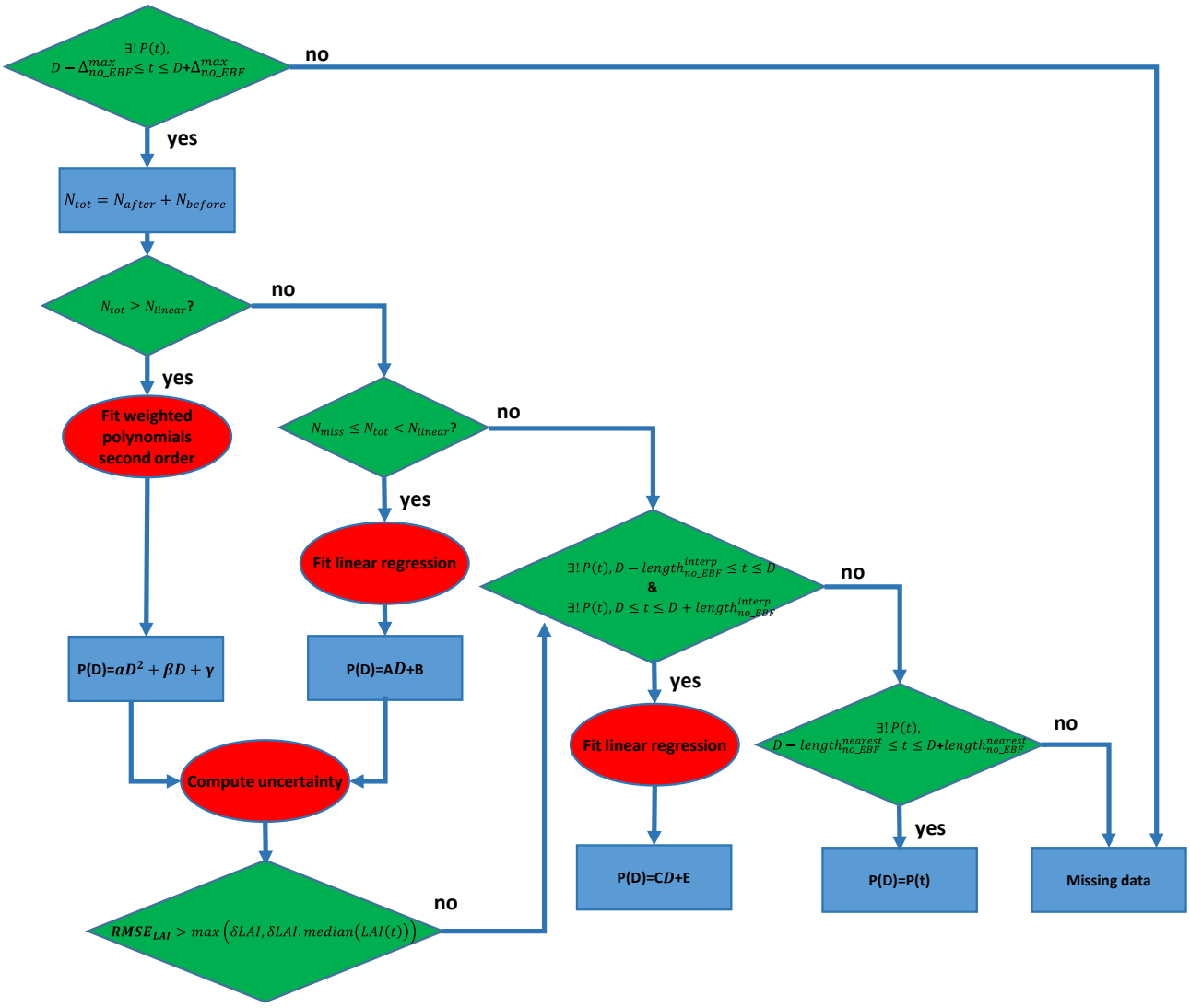


Figure 14: Flow chart showing the three cases considered for temporal compositing, smoothing and projection for non-Evergreen Broadleaf Forest situations. *t* represents the dates at which instantaneous products P are available. *D* indicates the current dekad at which the product is computed.

- ***If $N_{tot} \geq N_{linear}$, fitting a second-degree polynomial.*** If the number of available instantaneous estimates is higher than the minimum allowed number of observations for

fitting the polynomials, N_{linear}, a weighted second-degree polynomial fitting is applied. The weights are estimated in two iterations. The value of N_{linear} is set to $N_{linear} = 5$, as default value.

- For the first iteration, weights of available instantaneous estimates, w, are fixed to $w = 1.0$.
- For the second iteration, the weights, w, are computed according to the distance of the instantaneous estimates from the result of the first iteration. A sigmoidal function was considered for computing the weights w (Figure 15):

$$w = \frac{2}{1+\exp(-k*\delta)}$$

where δ is the difference between the instantaneous estimates and the first polynomial estimates, and $k = 2$ a slope parameter. Smaller weight values are considered when the instantaneous estimates are lower than the first polynomial fit since residual clouds or snow contamination generally lowers the value of the instantaneous product (Figure 16). Conversely, higher weights are considered for the larger δ values to better fit to the maximum values that are considered more reliable. This minimizes the possible underestimation due to the smoothing procedure observed over peaks in the first iteration, when all the weights w are set to 1.0.

The corresponding flag QC(7:6) in Table 8 is set to 00.

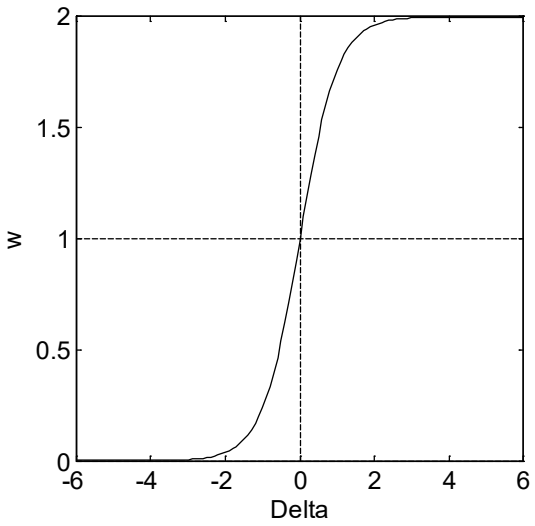


Figure 15: The weighting function used in the polynomial fitting. Delta represents the difference between the actual instantaneous data and the estimates from the first iteration of the polynomial. The slope *k* was set here to *k=2*.

- ***If*** $\boldsymbol{N_{miss} \leq N_{tot} < N_{linear}}$***, adjusting a linear fit.*** When less than N_{linear} instantaneous estimates are available in the compositing window, but more than N_{miss} observations are available within the compositing window, a simple linear fit is applied to provide a value for dekad D. The value of N_{miss} is set to $N_{miss} = 3$, as default value. The same principles are used as in the previous case, with one fit with $w = 1$ followed by one fit with the weights computed as previously using the function presented in Figure 15.

  The corresponding flag QC(7:6) in Table 8 is set to 10.

  Check if estimates are within the confidence interval: When used in projection mode for cases where little observations are available, the polynomial fit may lead to inconsistent estimates. These situations are identified based on the uncertainty of LAI (section 4.3.2.2.5, step B2.5). A threshold value is defined as the maximum value between an absolute and relative value: $max\left(\delta LAI, \delta LAI.median\left(LAI(t)\right)\right)$ where $\delta LAI = 0.35$ and $median(LAI(t))$ corresponds to the median value of instantaneous LAI estimates in the compositing window. In projection mode (RT0) when $N_{after} = 0$ or $N_{before} = 0$, if $RMSE_{LAI} > max\left(\delta LAI, \delta LAI.median\left(LAI(t)\right)\right)$, the product values are considered to be out of the confidence interval and recomputed as described here below.

- ***If*** $\boldsymbol{N_{tot} < N_{miss}}$, if less than N_{miss} instantaneous estimates are available in the compositing window, or $\boldsymbol{RMSE_{LAI} > max\left(\delta LAI, \delta LAI.median\left(LAI(t)\right)\right)}$, estimates out of confidence interval, then 3 cases must be considered:
  - Two observations, one before and one after the dekad are available within $\pm length_{no_EBF}^{interp}$ days (set to 15): then, a simple linear interpolation between those two instantaneous estimates is performed at the dekad date. The corresponding flag QC(7:6) in Table 8 is set to 01.
  - A single instantaneous estimate is available within $\pm length_{no_EBF}^{nearest}$ days (set to 5) around the dekad: then, the corresponding value is attributed to the dekad. The corresponding flag QC(7:6) in Table 8 is set to 11.
  - For the other cases, it results in missing data for that particular dekad. The corresponding flag QC(7:6) in Table 8 is set to 11 and the product value is set to 'missing' data (=255).

4.3.2.2.4 Output outlier rejection (Step B2.4)

To avoid possible artifacts introduced by divergences of the temporal filters, a post-filtering is applied to reject estimates out of the physical range of the variables.

The estimates lying outside the tolerance value range (tol_{LAI}, tol_{FAPAR}, tol_{FCOVER}) (Table 11) are rejected and set to missing values (255).

If one of the variables is considered invalid, all the three variables are rejected. This allows keeping a high level of consistency between the three products (LAI, FAPAR and FCover).

4.3.2.2.5 Computing the uncertainties (Step B2.5)

Two cases are considered depending on whether the product value is missing or not, and the number of available instantaneous valid estimates in the compositing window:

- If the product value does not correspond to a missing data and $N_{tot} \geq N_{miss}$, the root mean square error between the valid instantaneous product estimates $P(t)$ and the estimated product value from the polynomial fit at the dekad D, $P(D)$ is computed as:

$$RMSE = \sqrt{\frac{\sum_{t=1}^{N_{tot}} (P(t) - P(D))^2}{N_{tot}}}$$

  This RMSE metrics is a measure of the polynomial fit and a proxy of the uncertainty of the product value.
  In accordance with section 4.3.2.2.3, when $RMSE_{LAI} > max\left(\delta LAI, \delta LAI.median\left(LAI(t)\right)\right)$, the uncertainty for LAI, FAPAR and FCover are set to 255.

- If the product value is missing or $N_{tot} < N_{miss}$, the uncertainties values are not computed and they are set to 255.

4.4 LIMITATIONS OF THE ALGORITHM

The Version 2.0 algorithm for PROBA-V and Sentinel-3 300m products (and Sentinel-3 Version 1.1 300m products) capitalizes on the development and validation of PROBA-V Version 1.0 products and the use of neural networks. The PROBA-V Version 1.0 products used as target products for Version 2.0, the original CYCLOPES V3.1 and MODIS C5 used as target products for Version 1.0, the efficacy of the training process and the criteria used to define the input outliers will determine,

respectively, the magnitude and range of variation of the final products, their reliability and the definition domain used to remove contaminated values.

The Version 2.0 products are not accurate over the wetlands and can provide unrealistic values over water areas. This can be explained as a legacy effect since the Version 1.0 PROBA-V products used to train the Version 2.0 neural networks do not perform well over such surfaces since the original CYCLOPES (V3.1) and MODIS (C5) products were developed from physical radiative transfer models which are not applicable over heterogeneous surface mixed with water (Xu et al. 2020). In addition, these particular cases are not included in the training dataset used to generate CLMS global 300m products.

Some limitations of Version 2.0 300m products are associated to the reliability of the CCI-LC v2.0.7cds land cover even this auxiliary information was only used at the beginning of time series processing when no enough dekads are available for the estimation of EBF or nonEBF land cover class (section 4.3.2.1). The algorithm must optimally be applied in a sequential order starting with PROBA-V followed by Sentinel-3 processing. In this optimal case, the CCI-LC 2014 is used for the initialization EBF and nonEBF masks at the beginning of PROBA-V processing and for Sentinel-3 time series not external information is required but the EBF/nonEBF class resulted from Version 2.0 algorithm detection for the last dekad of PROBA-V processing is used. Note that because of the compositing period, the first dekad of the Sentinel-3 distributed products should preferentially be ~1 year after the date of the first daily input data ingested in the algorithm, more precisely at least 210 days after which corresponds to the maximum length of the half compositing window before the date being processed for EBF case.

The consistency across missions of long time series of 300m products highly depends on the temporal consistency of input TOC reflectance between PROBA-V and Sentinel-3 sensors and across the different processing chains (e. g. cloud-aerosol screening).

The efficiency of the temporal methods used in the composition of 300m products from instantaneous estimates depends on the level of noise and gaps in the time series. In this sense, the Collection 2 of PROBA-V data used as input in Version 2.0 is expected to improve the cloud screening of the earlier Collection 1 of PROBA-V used in Version 1.0 (https://proba-v.vgt.vito.be/en/quality/product-and-algorithm-information). The Sentinel-3 TOC reflectance includes a cloud screening based on IDEPIX [CGLOPS1_QAR_S3-CloudMask]. The final product is also dependent on the criteria used to filter the residual outliers, particularly for the tropical forests and high latitudes. Outlier rejection constitutes a critical step in the algorithm.

When used in projection mode for cases where little observations are available, the polynomial fit applied in the temporal smoothing and compositing may lead to inconsistent estimates. The time series in real time RT0 still show more noise compared to RT1, RT2 and the consolidated RT6 (see section 5.1) since, for these modes, more observations are available.

The approach used to process pixels flagged as EBFs constitutes an oversimplification of the reality because of the possible seasonality of evergreen broadleaf forests. The high uncertainty associated with the data due to poor atmospheric correction and very high cloud occurrence in equatorial and tropical latitudes prevented the extraction of meaningful seasonality at the resolution of the individual pixels (cf. site #19 in Figure 16, top).

The user should use the LAI, FAPAR, FCover 300m products with due attention to the quality flags values as well as the associated uncertainties, in particular for areas with long periods of cloudiness.

4.5 ALGORITHM DIFFERENCES WITH PREVIOUS VERSIONS OF 300M PRODUCTS

The main differences of algorithm Version 2.0 with algorithm Version 1.0 (PROBA-V products) and Version 1.1 (OLCI products) of algorithm refers to the input data and its preprocessing, i.e. atmospheric correction:

- Atmospheric correction: Both Version 1.0, Version 1.1 and Version 2.0. uses SMAC method for the atmospheric correction. The differences are that, in Version 1.0, the SMAC code is integrated in LAI/FAPAR/FCover retrieval algorithm which ingests TOA reflectance as input data while, in Version 1.1 and 2.0, the SMAC correction is applied in an earlier step of the CLMS processing chain and the LAI/FAPAR/FCover retrieval algorithm uses TOC reflectance as input data.
- Input reflectance
    - Instantaneous TOC reflectance products are used in Version 1.1 and Version 2.0 replacing the TOA daily synthesis reflectance used in Version 1.0.
    - Collection 2 of PROBA-V data are used in Version 2.0 while Collection 1 of PROBA-V was used in Version 1.0.
    - OLCI TOC reflectance Version 2.3 are used as input for products Version 2.0 while OLCI TOC reflectance Version 1.0 was used for product Version 1.1 up to 30 June 2022 and OLCI TOC reflectance Version 1.1 was used for product Version 1.1 from 1st July onwards.
- Auxiliary data
    - Atmospheric conditions: Information of the actual water vapor, ozone, pressure (derived from altitude) was required as input data in Version 1.0 to correct the TOA reflectance from atmospheric effects (except aerosol) using the SMAC4.0 model (Rahman and Dedieu 1994). These input data are not used in Version 1.1 and Version 2.0 since the algorithm ingests directly TOC reflectance that were already atmospherically corrected [CGLOPS1_ATBD_S3-AC-V1].

- Land cover: The CCI-LC land cover is used in Version 1.1 and Version 2.0 of algorithm instead of the GLOBCOVER (Defourny et al. 2009) map used in Version 1.0.

- Algorithm parameters: The parameters of the algorithm have been fine-tuned in Version 2.0 using actual PROBA-V and Sentinel-3 OLCI data. In particular, the following parameters have been modified:
  - Step A:
    - The parameters (weights and bias) of neural networks are specific for PROBA-V Collection 2 and Sentinel-3 OLCI since the training datasets (input and target variables) are different between Version 1.0, on one hand, and Version 1.1 and Version 2.0, in other hand. Note that the neural networks and their parameters are exactly the same for Version 1.1 and Version 2.0 Sentinel-3 products.
  - Step B:
    - N_{EBF} and $f_x(N_{tot})$ parameter in Step B1.2 (Section 4.3.2.1.2)
    - f_{EBF}^{min} parameter in Step B1.3 (Section 4.3.2.1.3)
    - $N_{outlier}^{min}$, $length_{no_EBF}^{outlier}$ parameter in Step B2.2 3 (Section 4.3.2.2.1)
    - $length_{no_EBF}^{min}$ parameter in Step B2.2 (Section 4.3.2.2.2)
    - δ_{LAI} parameter in Step B2.3 (Section 4.3.2.2.3)
    - $tol_{FAPAR}, tol_{FCOVER}$ in Step B2.4 (Section 4.3.2.2.4)
      Again, all these parameters in Step B, except $length_{no_EBF}^{outlier}$, are the same for Version 1.1 and Version 2.0 Sentinel-3 products.

Some modifications were also made in the Version 2.0 retrieval algorithm to address the limitations identified in Version 1.0 and Version 1.1 after the quality assessment exercises [CGLOPS1_QAR_LAI[FAPAR/FCOVER]300m-V1, CGLOPS1_QAR_LAI[FAPAR/FCOVER]300m-V1.1]:

- Step A5 (Section 4.3.1.5) was added in Version 1.1 and Version 2.0 for ensuring FAPAR-FCover consistency (FCover should not be larger than FAPAR, for a given pixel).

Some adaptations were also made in Version 1.1 and Version 2.0 due to operational constraints:

- Step B2.3 (Section 4.3.2.2.3): modification in the criteria to decide whether a composited dekadal product value was considered to be out of the confidence interval and recomputed.

4.6 RISK OF FAILURE AND MITIGATION MEASURES

In case one of the two Sentinel-3 sensors fails or its quality degrades, only one Sentinel-3 sensor will be used to continue the processing of LAI, FAPAR and FCover 300m products.

5 EVALUATION OF ALGORITHM PERFORMANCE

A preliminary evaluation of the Version 2.0 algorithm has been achieved, mostly based on the comparison of estimates in the different RT modes in the consolidated period, the comparison between Version 2.0 PROBA-V and Sentinel-3 300m products as well as on the comparison with previous Version 1.0 PROBA-V 300m products (Fuster et al. 2020) and MODIS C6 (Myneni et al. 2002; Yan et al. 2016a; Yan et al. 2016b) products. This was completed over the BELMANIP2 sites during the 2018-07-01 to 2019-06-30 period. For this assessment of algorithm performance, the comparison between Version 2.0 estimates with Version 1.1 Sentinel-3 products was not included because the algorithm remains the same.

The results of the quality assessment of the 300m LAI, FAPAR, FCover Version 1.0 PROBA-V products are presented in [CGLOPS1_QAR_LAI[FAPAR/FCOVER]300m-V1]. A full quality assessment of Version 1.1 Sentinel-3 LAI, FAPAR and FCover 300m products is reported in the validation report [CGLOPS1_QAR_LAI[FAPAR/FCOVER]300m-V1.1]. It includes the accuracy assessment of Version 1.1 Sentinel-3 products based on the comparison with ground data and an exhaustive validation of products based on user requirements. However, a reanalysis is required because a geolocation issue was identified in the preprocessing of Sentinel-3 which also affected the LAI, FAPAR and FCover products. Then, the quality assessment needs to be extended to Version 2.0 PROBA-V and Sentinel-3 products including the comparison with Version 1.0 and Version 1.1 products.

5.1 COMPARISON OF RT MODES OF VERSION 2.0 SENTINEL-3 PRODUCTS

Detailed investigation of the temporal profiles of RT Version 2.0 Sentinel-3 products (Figure 16) shows that after 2 dekads (RT2) the differences with the consolidated RT6 estimates are marginal. Conversely, the first real time estimates (RT0) with data available only in the past shows some discrepancies and higher instability compared with the final RT6 series, in particular for situations with few observations and scattering in the instantaneous estimates.

Results in Figure 17, Figure 18 and Figure 19 confirm a rapid convergence for the intermediate LAI, FAPAR and FCover solutions towards RT6 which show a similar pattern after two dekads (RT2).

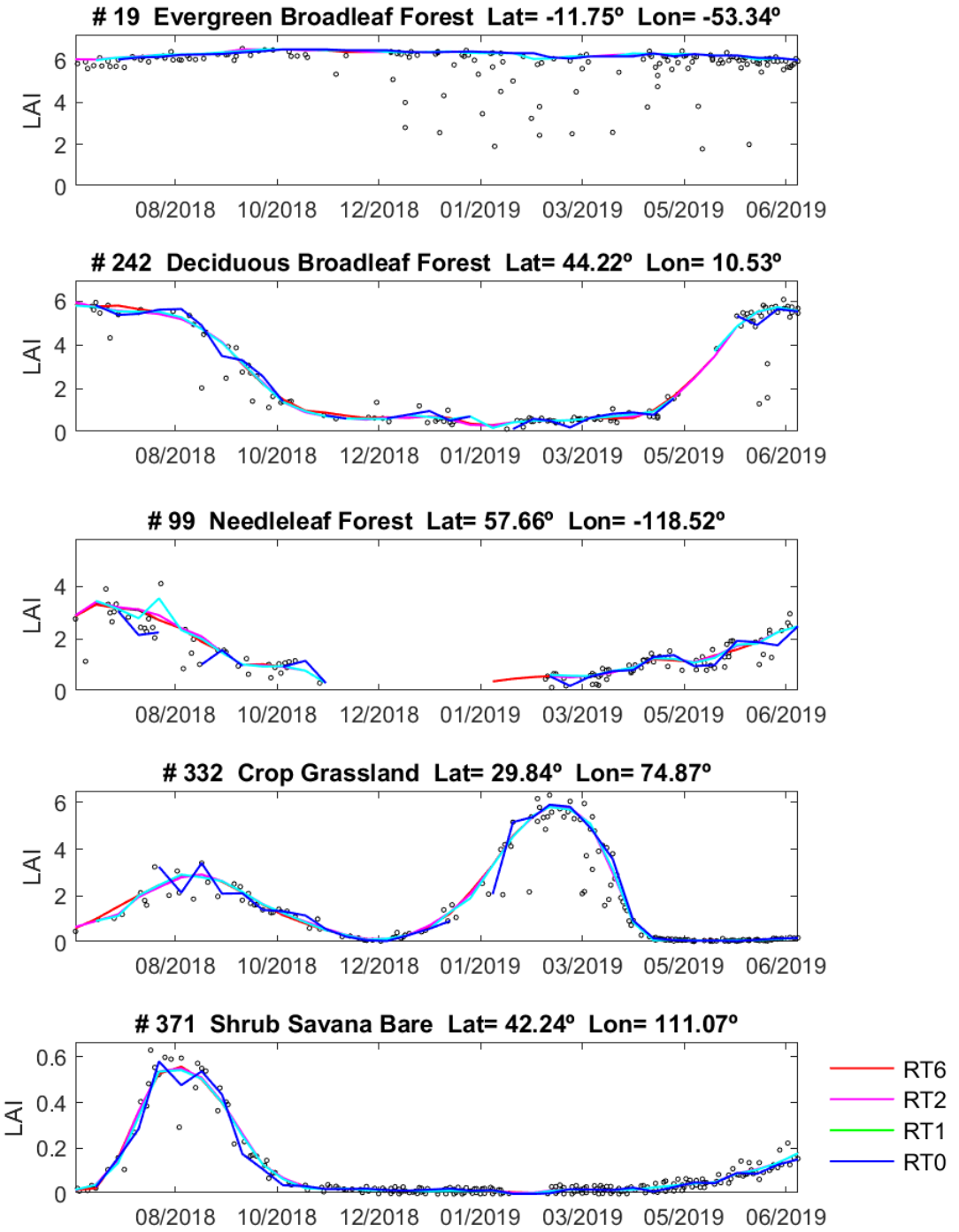


Figure 16: Temporal profiles of Version 2.0 RT (for RT0, RT1, RT2 and RT6) Sentinel-3 LAI products over several BELMANIP2 sites for the period from 2018-07-01 to 2019-06-30. The points represent the instantaneous LAI estimates. The title of each temporal profile indicates the BELMANIP2 site number, the CCI-LC biome class, the latitude and longitude in degrees.

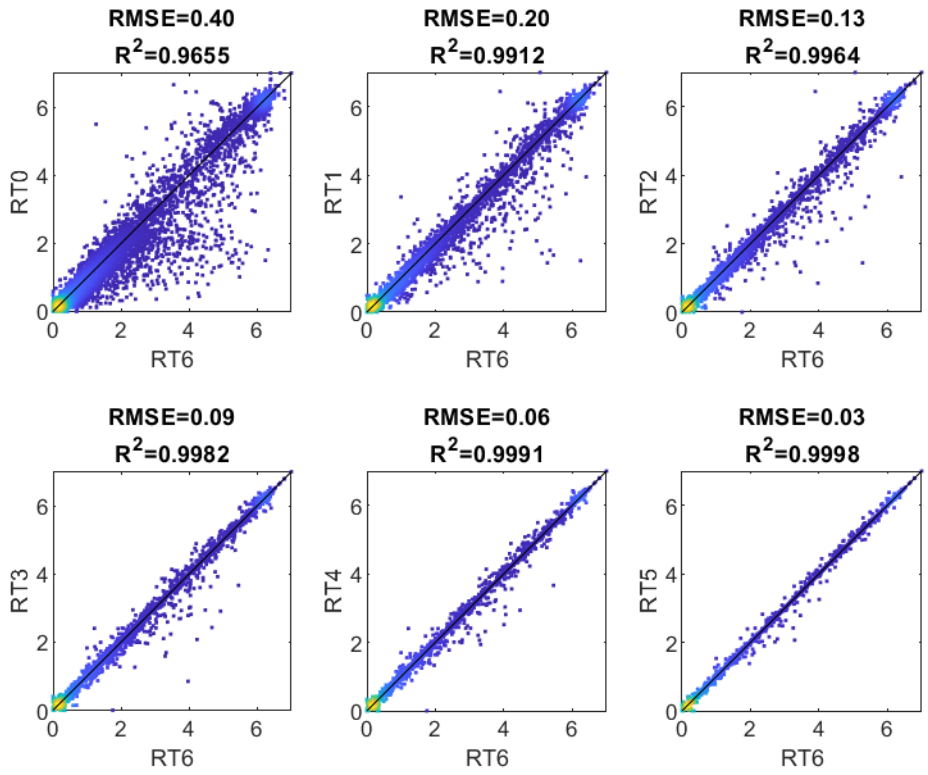


Figure 17: Comparison of the intermediate RT (from RT0 to RT5) and the consolidated RT6 modes of Version 2.0 Sentinel-3 LAI 300m products over the BELMANIP2 sites for the 2018-07-01 to 2019-06-30 period. The yellow tones correspond to more density of points. The root mean square error (RMSE) and the coefficient of determination (R^2) are indicated.

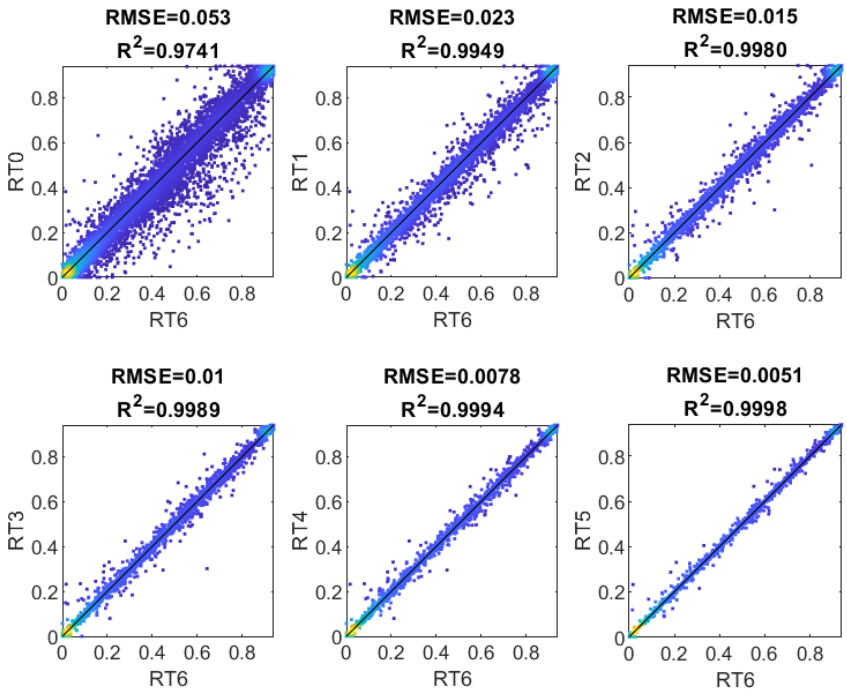


Figure 18: Idem as Figure 17 for FAPAR.

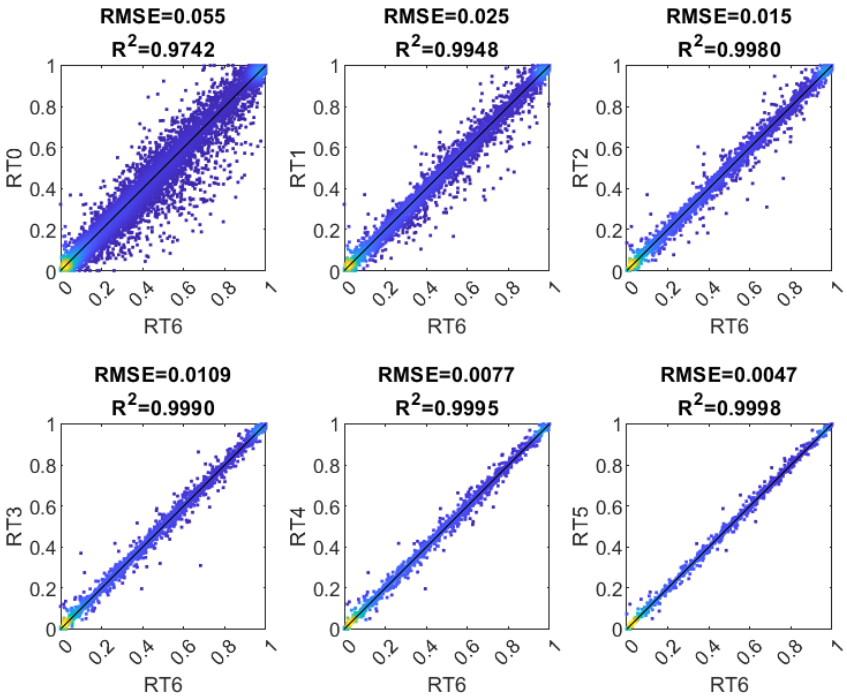


Figure 19: Idem as Figure 17 for FCover.

An exhaustive sensitivity analysis of RT (from the initial solution RT0 to the consolidated product RT6) to the number of observations and data noise is provided in Figure 20, Figure 21 and Figure 22. The RMSE between intermediate RT solutions and the consolidated RT6 estimates of V2.0 300m products linearly increases with the amount of noise in the data (computed as the RMSE of instantaneous estimates compared to RT6). The slope of this linear relationship is higher for the first RT updates with the maximum value for RT0. The slopes are lower than 1 in all RT modes which means higher agreement with the consolidated RT6 for the intermediate RT products compared to the instantaneous estimates. The RMSE between RT-x and RT6 shows a rapid exponential decay with the number of available instantaneous estimates (Figure 20b, Figure 21b and Figure 22b). It is almost zero for 15 valid estimates before the dekad being processed. A rapid convergence is observed for the intermediate solutions, which show a similar pattern after two dekads (RT2). In all cases, including the situations when few valid instantaneous estimates are available and/or data are associated to a high level of noise, the discrepancies between RT-x and RT6 in terms of RMSE are <0.5 for LAI and <0.08 for FAPAR and FCover.

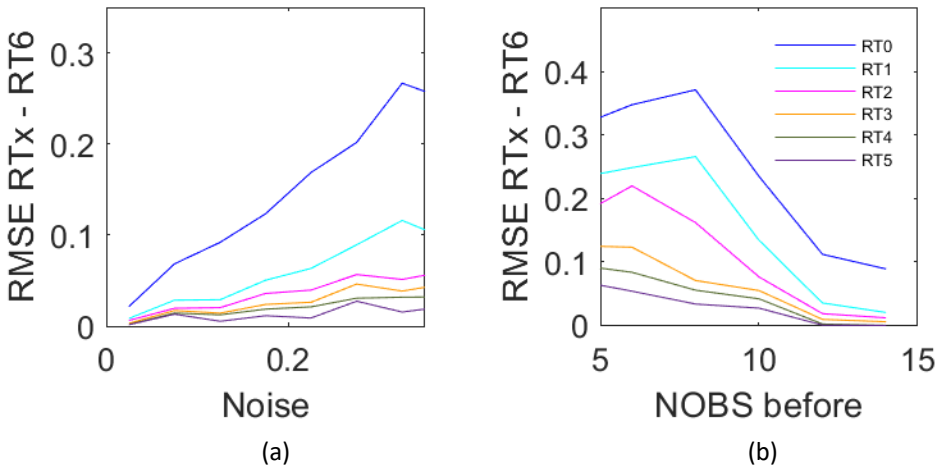


Figure 20: RMSE between the intermediate RT (for RT0 to RT5) and the consolidated RT6 modes of Version 2.0 Sentinel-3 LAI 300m products as a function of (a) the noise in the data computed as the RMSE between the instantaneous estimates and the RT6 product and (b) the number of available instantaneous estimates (NOBS) before the date of the dekad being processed.

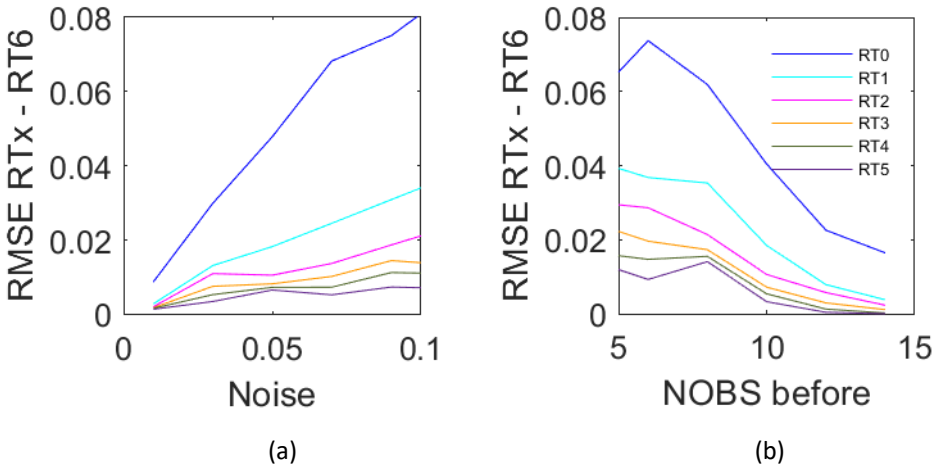


Figure 21: Idem as Figure 20 for FAPAR.

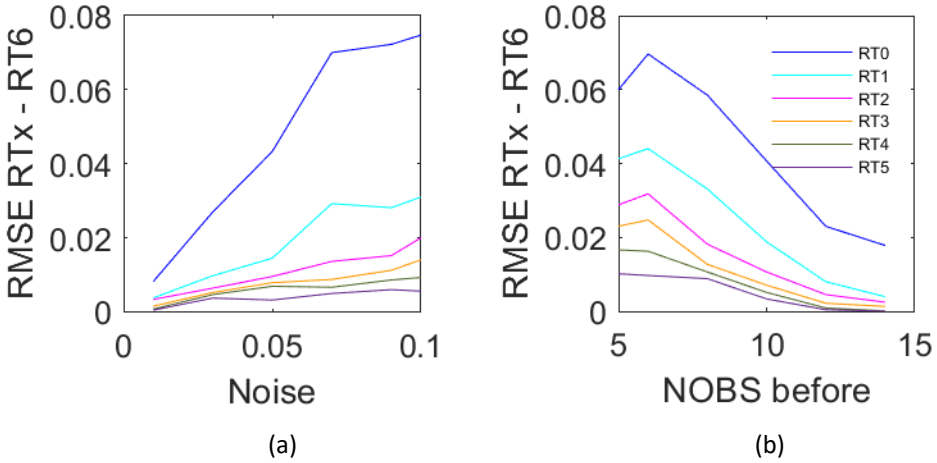


Figure 22: Idem as Figure 20 for FCover.

The continuity of time series is evaluated by the fraction of valid data per biome types (Figure 23). An improvement in continuity is observed from RT0 to RT1 as expected since RT1 takes advantage of the instantaneous estimates available within 10 days after the dekad being processed while RT0 uses only the observations before this date. A slight improvement in terms of continuity is observed after RT1. Identical patterns per biome are observed, but with a lower fraction of valid data for needleleaf forests mostly located at high latitudes areas affected by snow cover and poor illumination conditions. The higher continuity of evergreen broadleaf time series compared to the needleleaf forests is due to the different composition scheme applied for EBFs in step B1 with a longer composition period.

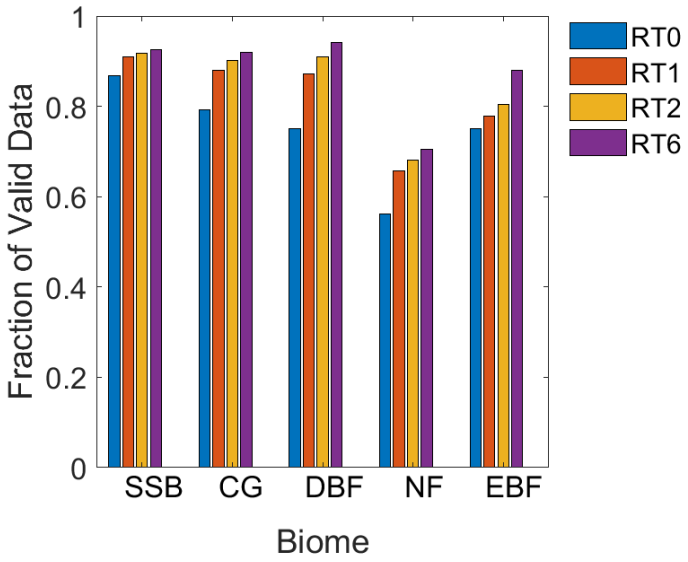


Figure 23: Fraction of valid data for the different RT modes of Version 2.0 Sentinel-3 300m products per biome. The biome classes are derived from the CCI-LC land cover: Shrubs/Savana/Bare soil (SSB), Crops and Grassland (CG), Deciduous Broadleaf Forests (DBF), Needleleaf Forest (NF), and Evergreen Broadleaf Forest (EBF). Evaluation over the BELMANIP2 sites for the 2018-07-01 to 2019-06-30 period.

The smoothness of the time series was evaluated based on the absolute value of the difference, δ, between the product value at date *t* and the mean value between the two closest bracketing dates in a maximum Δt period of 60 days: $\delta LAI = |1/2(LAI(t+\Delta t) + LAI(t-\Delta t)) - LAI(t)|$ for the case of LAI (Verger et al. 2011b). The smoother the temporal evolution, the smaller the δ difference should be. The histogram of δ (Figure 24) shows that the time series of different RT modes of V2.0 300m products are very smooth with differences <0.25 LAI and <0.05 FAPAR and FCover for most of cases. RT0 shows the shakier temporal evolution while an improvement in terms of smoothness is observed one dekad after (compare histograms of RT1 and RT0) which rapidly converges towards RT6 distributions.

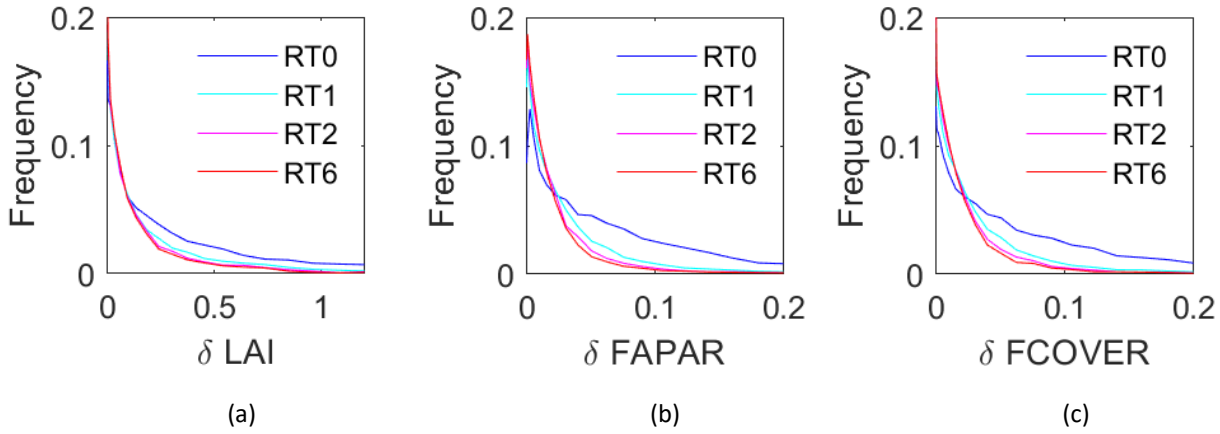


Figure 24: Histogram of the δ absolute difference representing temporal smoothness for the different RT modes of Version 2.0 Sentinel-3 (a) LAI, (b) FAPAR and (c) FCover 300m products. Evaluation over the BELMANIP2 sites for the 2018-07-01 to 2019-06-30 period.

5.2 CONSISTENCY BETWEEN LAI, FAPAR AND FCOVER

In terms of the internal consistency between LAI, FAPAR and FCover variables (Figure 25, Figure 26), Version 2.0 PROBA-V and Sentinel-3 products show very similar relationships and both products improve Version 1.0 PROBA-V 300m products:

- The LAI-FAPAR relationship (Figure 25) for V1.0 and V2.0 of products show an exponential relationship as expected (Myneni and Williams 1994). The comparison of LAI-FAPAR for V2.0 shows high consistency for all the range of values with non-artifacts introduced by the use of two different processing schemes: one for EBFs and one for non EBFs classes. The V2.0 LAI-FAPAR relationship is very similar to the V1.0 one which makes no distinction per classes.
- The FAPAR-FCover relationship (Figure 26) is more robust for V2.0 than for V1.0 which shows some inconsistencies for FCover>0.5. Indeed, FCover (gap fraction at nadir) should always be lower than FAPAR that can be approximated by the FIPAR (gap fraction in the sun direction). This is the case for V2.0 as a consequence of the theoretical constrain imposed in Step A5 (section 4.3.1.5) of V2.0 algorithm: FCover < FAPAR/0.94. However, this is not the case for V1.0. The inconsistencies observed in V1.0 can be corrected forcing FCover V1.0 product to be lower than normalized FAPAR: FCover < FAPAR/0.94. The FAPAR-FCover relationship for V1.0-Corr. PROBA-V products is similar to the V2.0 PROBA-V and V2.0 Sentinel-3 ones.

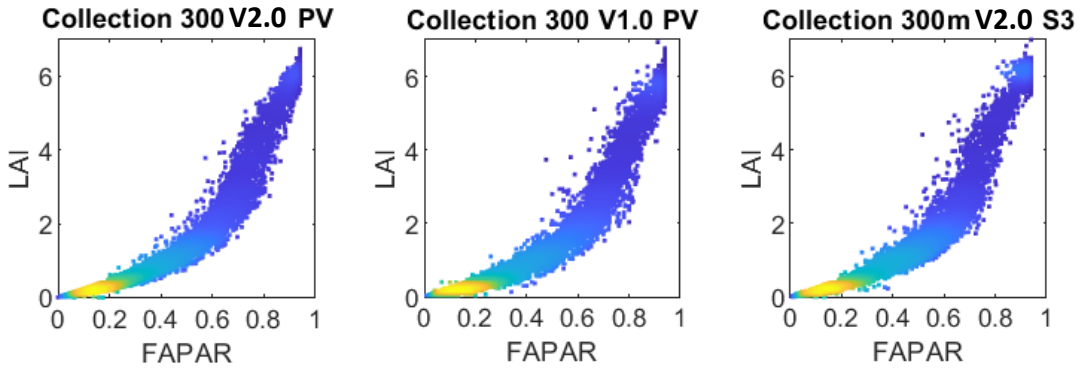


Figure 25: Comparison between LAI-FAPAR for V1.0 and V2.0 of 300m PROBA-V (PV) and Sentinel-3 (S3) products over the BELMANIP2 sites for the 2018-07-01 to 2019-06-30 period (number of samples, n=9002). The yellow tones correspond to more density of points.

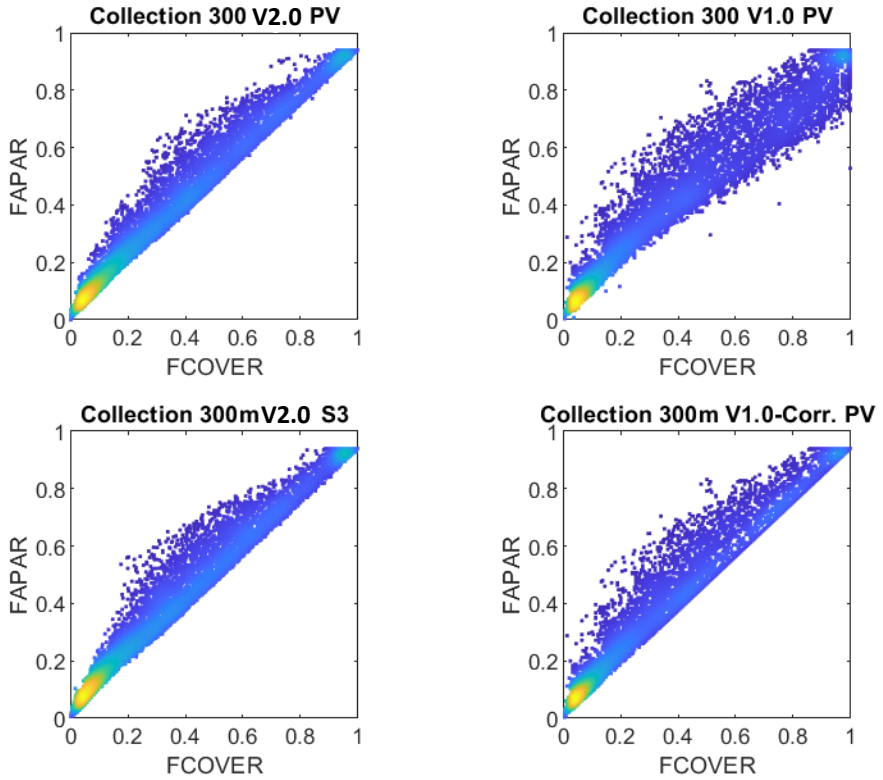


Figure 26: Comparison between FAPAR-FCover for V1.0 and V2.0 of 300m PROBA-V (PV) and Sentinel-3 (S3) products over the BELMANIP2 sites for the 2018-07-01 to 2019-06-30 period (number of samples, n=9002). V1.0-Corr. corresponds to the corrected V1.0 300m PROBA-V forcing FCover < FAPAR/0.94. The yellow tones correspond to more density of points.

5.3 COMPARISON BETWEEN 300M PROBA-V AND SENTINEL-3 VERSION 2.0 AND WITH PROBA-V VERSION 1.0 AND MODIS C6

Sentinel-3 Version 2.0 and PROBA-V Version 1.0 show an overall RMSE 0.43 LAI with most of differences coming from random noise mostly due to the differences in the observations from the two different sensors (Figure 27a). Similar overall performance (RMSE=0.43) is observed for the comparison of V2.0 and V1.0 PROBA-V with however some systematic differences for the highest LAI values corresponding to Evergreen Broadleaf Forests (EBFs) (Figure 27b, Figure 28). These slight differences for EBFs are also observed between Sentinel-3 V2.0 and PROBA-V V1.0 (Figure 27c, Figure 28). This can partially be explained due to the differences in Step A of V2.0 and V1.0 algorithms: a specific neural network for EBFs is applied in V2.0 for the generation of instantaneous estimates while the same NNT across biomes is applied in V1.0. V2.0 is closer to MODIS LAI for EBFs (Figure 28). The comparison between Version 2.0 and MODIS C6 LAI products shows a high agreement for the low and high values but V2.0 provides higher intermediate LAI values than MODIS (Figure 27d-e) particularly for deciduous broadleaf forests (Figure 28). The high scattering observed for intermediate and high values is associated to the noise in the products and the differences in the pre-processing (cloud screening, atmospheric correction) and retrieval algorithms for MODIS (Myneni et al. 2002) and V2.0. Only MODIS C6 values retained with the main algorithm in optimal conditions were used for the validation, i. e. quality flag of MCD15A3H product with Bit0: Good quality; Bit2: Detectors apparently fine for up to 50% of channels 1,2; Bit3-4: Significant clouds not present; Bit5-7: Main (RT) method used (with or without saturation).

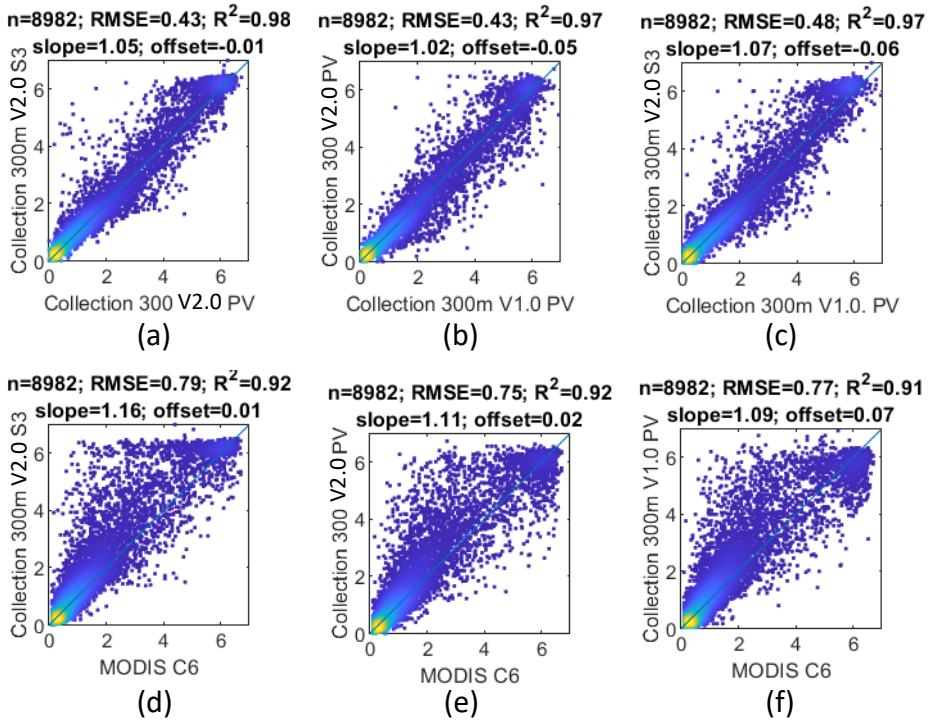


Figure 27: Comparison between V2.0 Sentinel-3 and PROBA-V (a), V2.0 and V1.0 PROBA-V (b), V2.0 Sentinel-3 and V1.0 PROBA-V (c), and and MODIS C6 (d,e,f) LAI products over the BELMANIP2 sites for the period from 2018-07-01 to 2019-06-30. The yellow tones correspond to more density of points. The number of samples (n), the root mean square error (RMSE), the coefficient of determination (R^2), the slope and the intercept of the linear regression are indicated.

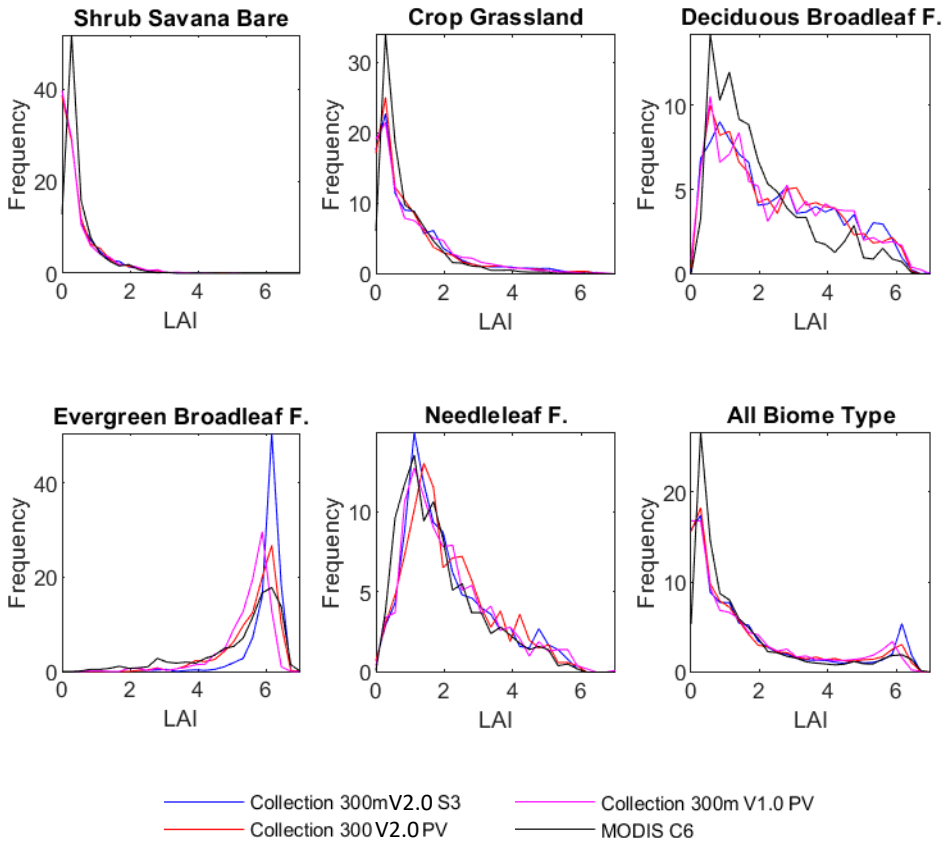


Figure 28: Distribution of V2.0 PROBA-V (PV) and Sentinel-3 (S3), V1.0 PROBA-V, and MODIS C6 LAI products per CCI-LC biome type over the BELMANIP2 sites for the period from 2018-07-01 to 2019-06-30. The bin size of the histograms is 0.25 LAI.

For FAPAR, variable less affected by noise in the data, V2.0 Sentinel-3 and PROBA-V products shows a high agreement for the entire range of FAPAR values (Figure 29a) and biomes (Figure 30). Similarly, V2.0 and V1.0 FCover products highly agree (Figure 29b-c). V2.0 and V1.0 FCover shows slightly higher differences with MODIS C6 (Figure 29d-f) and particularly for the low FAPAR values that are recognized to be overestimated by MODIS (Camacho et al. 2013). The distributions of FAPAR (Figure 30) show this higher discrepancy of MODIS with a positive bias as compared to CLMS 300m products for low FAPAR values. Again, CLMS 300m FAPAR products show higher intermediate-high values than MODIS for Deciduous Broadleaf Forests. Some discrepancies are also observed for the high FAPAR values and, in particular, for EBF biome where CLMS 300m FAPAR products shows distributions of values closer at the theoretical maximum for FAPAR (0.94) as compared to MODIS.

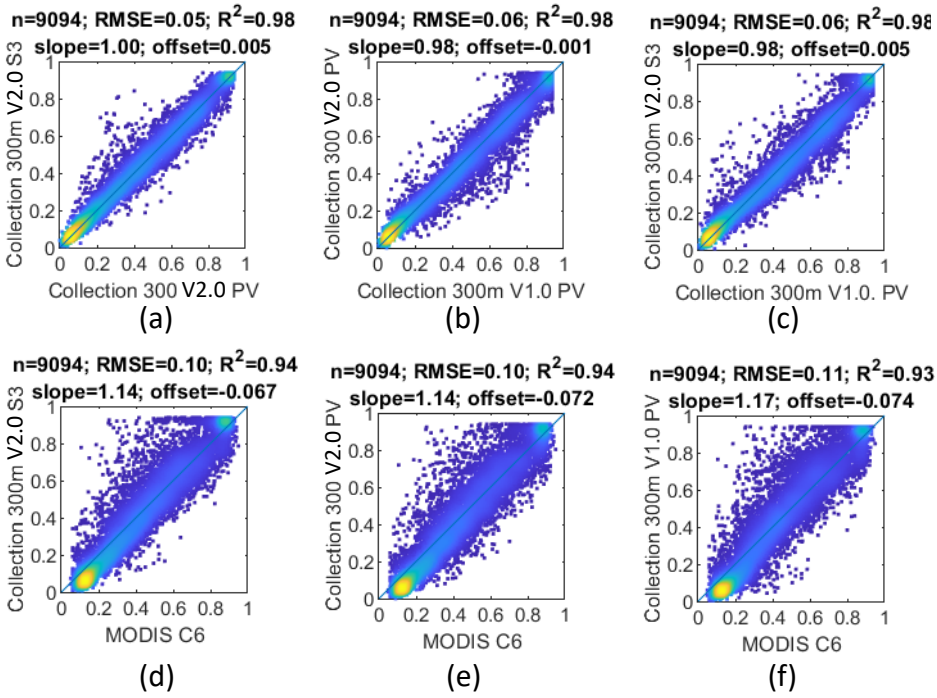


Figure 29: Comparison between V2.0 Sentinel-3 and PROBA-V (a), V2.0 and V1.0 PROBA-V (b), V2.0 Sentinel-3 and V1.0 PROBA-V (c), and 300m and MODIS C6 (d,e,f) FAPAR products over the BELMANIP2 sites for the period from 2018-07-01 to 2019-06-30. The yellow tones correspond to more density of points. The number of samples (n), the root mean square error (RMSE), the coefficient of determination (R^2), the slope and the intercept of the linear regression are indicated.

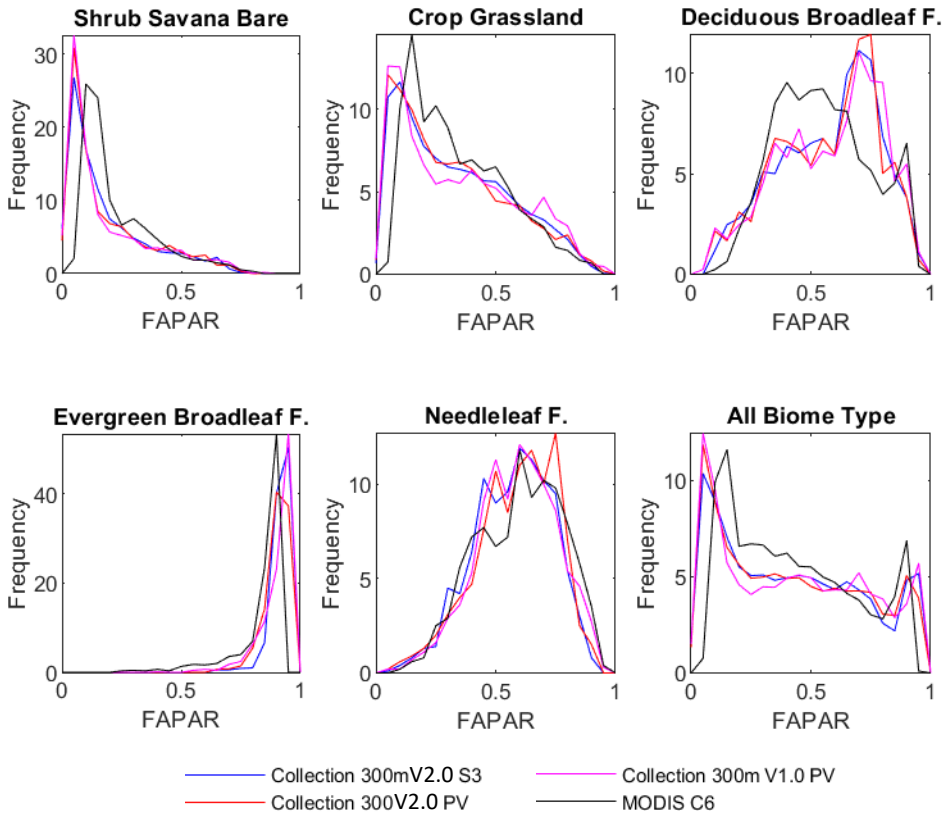


Figure 30: Distribution of V2.0 PROBA-V and Sentinel-3, V1.0 PROBA-V, and MODIS C6 FAPAR products per biome type over the BELMANIP2 sites for the period from 2018-07-01 to 2019-06-30. The bin size of the histograms is 0.05 FAPAR.

V2.0 Sentinel 3 and PROBA-V FCover products highly agree (RMSE=0.05; Figure 31a). The comparison of V2.0 with V1.0 PROBA-V FCover, shows an overall performance of 0.09 FCover in terms of RMSE for both V2.0. PROBA-V (Figure 31b) and Sentinel-3 (Figure 31c) products. A high agreement between V2.0 and V1.0 products is observed for the FCover values lower than 0.5. For FCover>0.5, V1.0 systematically provides higher values than V2.0, particularly for the non EBFs (Figure 32). Note that the direct validation based on the comparison with ground measurements revealed that V1.0 PROBA-V overestimates high FCover values [CGLOPS1_QAR_LAI[FAPAR/FCOVER]300m-V1] (Fuster et al. 2020). Indeed, V2.0 aimed to reduce this apparent overestimation for intermediate-high FCover values and the FAPAR-FCover inconsistencies. This positive bias in V1.0 FCover can be also corrected forcing FCover<FAPAR/0.94 (Figure 31d). Both V2.0 PROBA-V (Figure 31e) and Sentinel-3 (Figure 31f) show a good agreement with V1.0-Corr.

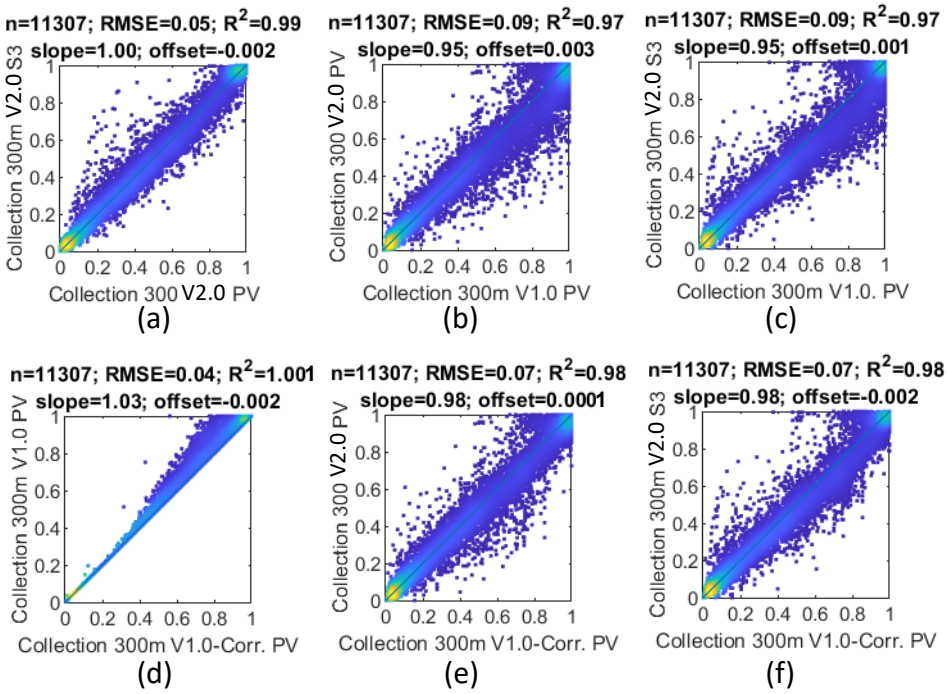


Figure 31: Comparison between V2.0 Sentinel-3 and PROBA-V (a), V2.0 and V1.0 PROBA-V (b), V2.0 Sentinel-3 and V1.0 PROBA-V (c), V1.0 and corrected V1.0 forcing FCover<FAPAR/0.94 (V1.0-Corr.) (d), V2.0 and V1.0-Corr. PROBA-V (e), and V2.0 Sentinel-3 and V1.0-Corr (f) FCover products over the BELMANIP2 sites for the period from 2018-07-01 to 2019-06-30. The yellow tones correspond to more density of points. The number of samples (n), the root mean square error (RMSE), the coefficient of determination (R^2), the slope and the intercept of the linear regression are indicated.

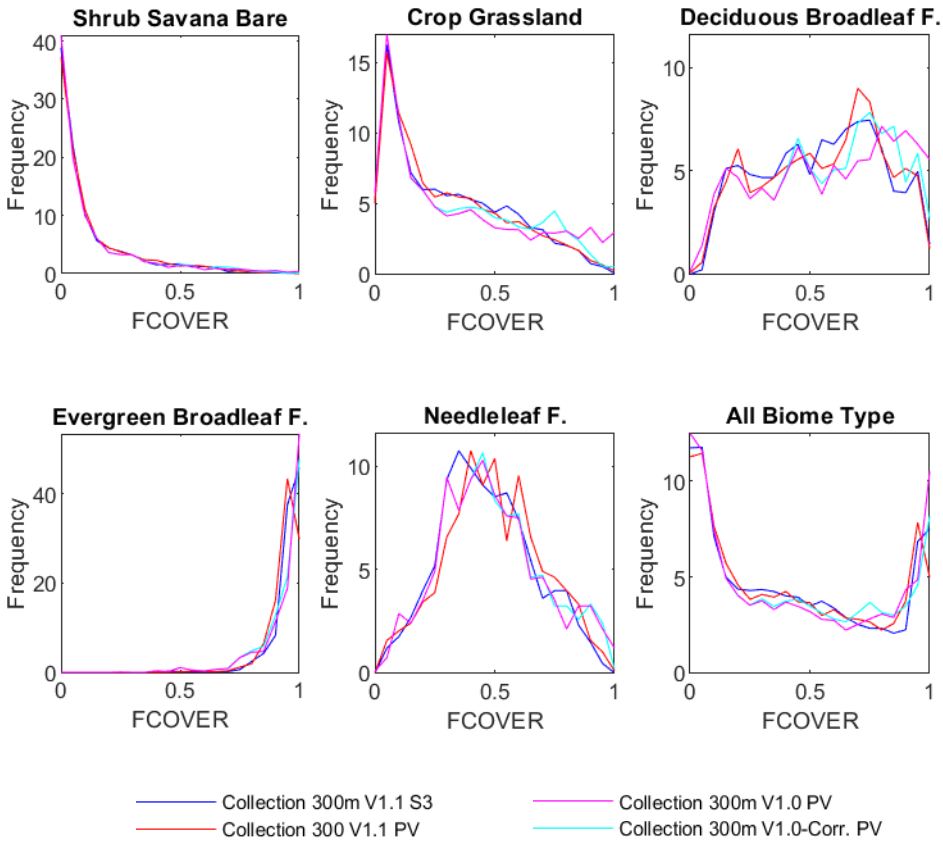


Figure 32: Distribution of V2.0 PROBA-V, V2.0 Sentinel-3 and V1.0 PROBA-V FCover products as well as corrected V1.0 forcing FCover<FAPAR/0.94 (V1.0-Corr.). Assessment per CCI-LC biome type over the 445 BELMANIP2 sites for the period from 2018-07-01 to 2019-06-30. The bin size of the histograms is 0.05 FCover.

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ANNEX 1: NEURAL NETWORK CALIBRATION

The V2.0 retrieval algorithm is based on the application of neural networks for each variable (LAI, FAPAR, FCover) and land cover class (EBF and nonEBF). The neural networks are also sensor specific: one set of neural networks is calibrated for PROBA-V and other for Sentinel-3. Then the combination of three variables (LAI, FAPAR, FCOVER) per two landcover vegetation classes (EBF and nonEBF) per two sensors (PROBA-V and Sentinel-3) resulted in twelve neural networks.

The neural networks were calibrated to mimic PROBA-V V1.0 LAI, FAPAR and FCover 300m products (Figure 33). The input data for PROBA-V and Sentinel-3 neural networks was PROBA-V C2 TOC reflectance [PROBAV_PUM_C2] and Sentinel-3/OLCI TOC reflectance V1.1 [CGLOPS1_PUM_S3-TOC-Reflectance300m-V2.3], respectively. The PROBA-V V1.0 LAI, FAPAR and FCover 300m products were used as the target variables instead of Sentinel-3 V1.1 products because (i) the Sentinel-3 V1.1 products were already estimated using neural networks calibrated on PROBA-V V1.0, (ii) There is no need to change the neural networks for Sentinel-3 products: the neural networks used for V2.0 Sentinel-3 products are exactly the same (same architecture and parameters) as the ones used for V1.1 Sentinel-3 products, (iii) Sentinel-3 V1.1 products, until 30 June 2022, are affected by geolocation issues.

The calibration process consists mainly of four steps:

- Generating the training database
- Generating the definition domain
- Defining the neural network architecture
- Calibrating the network.

These several steps have been investigated in detail by Verger et al. (2011a).

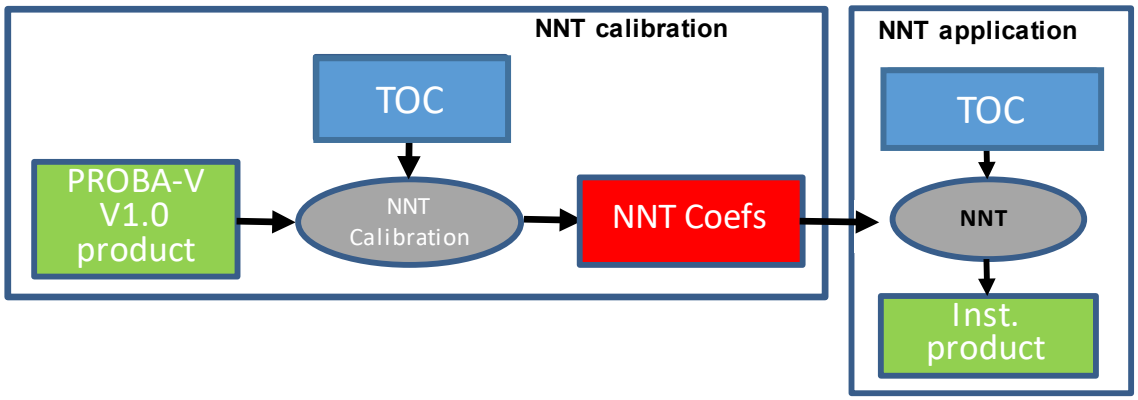


Figure 33: Schematic description of the principles used for V2.0 neural networks calibration and application. TOC refers to the input top of the canopy reflectance from PROBA-V or Sentinel-3.

1. Generation of the training database

The generation of the training database corresponds to the most critical issue to be solved. It should be based on accurate and representative input TOC reflectance and output LAI FAPAR and FCover variables.

Since the objective was to achieve consistency with PROBA-V V1.0 LAI, FAPAR and FCover 300m products, the neural networks were trained using concurrent input TOC reflectance (from either PROBA-V C2 or Sentinel-3/OLCI V1.1) and PROBA-V V1.0 LAI, FAPAR and FCover products over one year overlapping period. We used only one year because of operational constraints. Preparing the training dataset was a very time/computational consuming task particularly for Sentinel-3 because TOA S3 reflectance needed to be processed at the global scale to generate the TOC reflectance dataset. The PROBA-V and Sentinel-3 overlapping period from 1/07/2018 to 30/06/2019 was selected to have access to both Sentinel-3A and -3B data and, at the same time, to prevent from possible artifacts in PROBA-V due to sensor degradation at the end of PROBA-V mission.

The neural networks were calibrated over the BELMANIP2.1 set of sites (Figure 34). BELMANIP2.1 (http://calvalportal.ceos.org/web/olive/site-description) is an update of the original CEOS-BELMANIP1 ensemble of sites used for the calibration and validation of land products from medium resolution sensors (Baret et al. 2006). This set of 445 sites was defined to represent the possible range of vegetation types and conditions over the Earth while showing little topography and good level of homogeneity. The land cover homogeneity of each site was double checked using the GLOBCOVER map, and the Google Earth engine (Weiss et al. 2014).

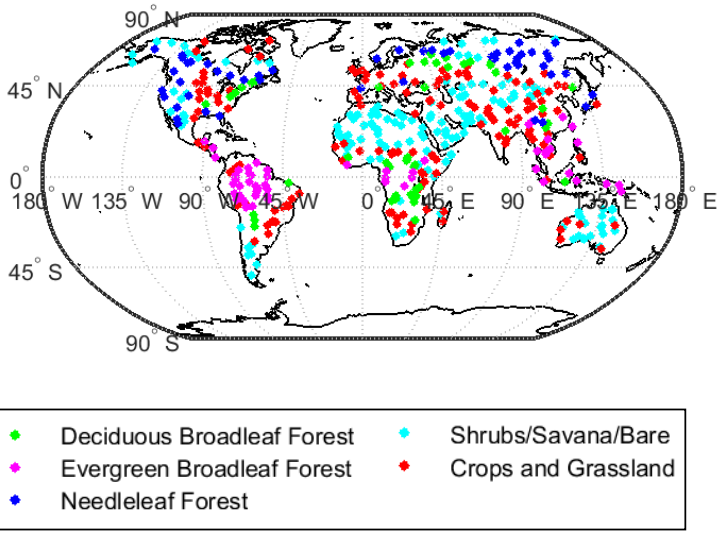


Figure 34. Location of BELMANIP2.1 sites per CCI-LC biome type.

The inputs of the neural networks are:

- TOC reflectance (the PROBA-V neural networks ingest TOC reflectance in 3 spectral bands: B0, B2 and B3, while the Sentinel-3 neural networks ingest TOC reflectance in 12 OLCI spectral bands: Oa4-12 and Oa16-18),
- the cosine of the view zenith angle (cos(VZA)),
- the cosine of the sun zenith angle (cos(SZA)),
- the cosine of the relative azimuth angle (cos(SAA-VAA)),

The outputs of the neural networks for the calibration are the final PROBA-V V1.0 LAI, FAPAR and FCover 300m products after consolidation [ImagineS_RP2.1_ATBD-LAI300m_I1.73]. To improve the FAPAR-FCover consistency (Figure 26), when preparing the training dataset, the theoretical constrain FCover < FAPAR/0.94 was imposed and the original PROBA-V V1.0 FCover data were transformed and decreased its value to FAPAR/0.94 if higher than it.

As the neural network developed for producing PROBA-V V1.0 depends on CYCLOPES (V3.1) and MODIS (C5), Sentinel-3 V2.0 may also depend on quality of both target products (or legacy effect).

The performances of the neural networks technique are closely linked to the way the learning database is generated. To generate the training data base, all the inputs (reflectance and angles) and outputs (products) should share the same spatial and temporal resolution. To minimize the effects of geometrical uncertainties and differences between products, a 3x3 pixel area centered over the BELMANIP2.1 sites was considered. The selection process assumes that the 3x3 pixels around the sites are homogeneous, which is mainly the case for the BELMANIP2.1 sites (Weiss et al. 2014). The input and output data were filtered and only high-quality pixels were retained for the calibration of neural networks:

➢ *Inputs*

- The input TOC data were filtered based on their quality status map. Only pixels with input TOC data having good radiometric quality for all the selected spectral bands, located over land, not covered by ice, cloud or snow were used:

  - For PROBA-V C2 TOC data, the QC plane (Table 2) is first used to keep only the best quality pixels i.e. pixels with status map value equal to 248 = 11111000.

  - For Sentinel-3 V1.1 TOC data:

    - quality_flags layer (ACRI-ST, 2017):
      - ‘saturated_Oa*’ for bands Oa4-12 and Oa16-18, exclude when raised
    - pixel_classif_flags (see Figure 1 in [CGLOPS1_QAR_S3-CloudMask]):

- IDEPIX_LAND: include when raised
- IDEPIX_CLOUD: exclude when raised
- IDEPIX_CLOUD_AMBIGUOUS: exclude when raised
- IDEPIX_CLOUD_BUFFER: exclude when raised
- IDEPIX_CLOUD_SHADOW: exclude when raised
- IDEPIX_SNOW_ICE: exclude when raised

- For each date, input TOC reflectances and angles were averaged in the 3x3 window.

➢ *Outputs*

A dedicated selection process was then applied to get as consistent as possible products regarding the possible contamination by clouds. It is achieved for each date where an instantaneous Sentinel-3 (PROBA-V) TOC product is available through three steps:

- Selection of PROBA-V LAI, FAPAR and FCover products: all the PROBA-V V1.0 products available within ±10 days around the date of the instantaneous TOC data are gathered over the 3x3 pixels.
- The 70 percentile values of LAI, FAPAR and FCover are computed over the selection of PROBA-V V1.0 products. The 70-percentile value is selected to reduce the occurrence of cloud and atmosphere artifacts. If no valid products are available, this results in a missing value. This step is applied only when the site is not considered as a bare soil based on GEOCLIM (Verger et al. 2015). A site is identified as bare soil if the 90-percentile of the GEOCLIM LAI is less than 0.05. For these bare soil sites (a total of 58), the variables were set to 0.
- The remaining outliers in the training database were removed based on fitted upper and lower exponential relationships between LAI and NDVI (Figure 35).
    - For the PROBA-V neural networks, the data that are selected for the learning database follow:

$$\begin{cases} LAI_{PV} \leq 0.260 \exp(3.239\, NDVI_{PV}) + 8.618 \times 10^{-3} \exp(7.705\, NDVI_{PV}) \\ LAI_{PV} \geq 0.054 \exp(5.177\, NDVI_{PV}) - 0.295 \exp(-2.571 \times NDVI_{PV}) \end{cases}$$

    - For the Sentinel-3 neural networks, the data that are selected for the learning database follow:

$$\begin{cases} LAI_{PV} \leq 2.527 \times 10^{-12} \exp(37.260\, NDVI_{S3}) + 0.228 \exp(3.707\, NDVI_{S3}) \\ LAI_{PV} \geq 0.032 \exp(5.281\, NDVI_{S3}) - 16.550 \exp(-9.202 \times NDVI_{S3}) \end{cases}$$

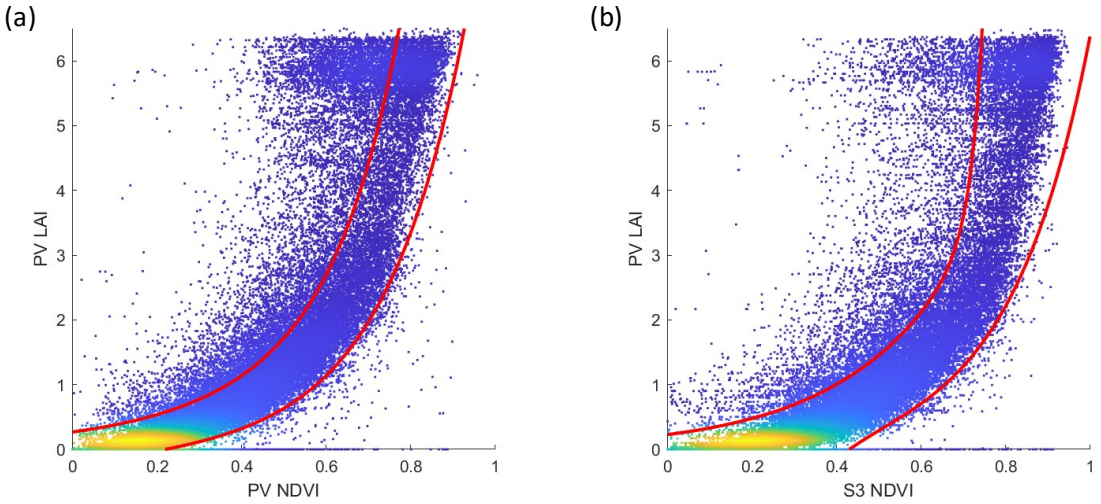


Figure 35: Filtering of the training dataset based on the LAI-NDVI relationship for (a) PROBA-V neural networks, i.e. PROBA-V (PV) V1.0 LAI and NDVI derived from PV Collection 2 TOC reflectance data and (b) Sentinel-3 neural networks, i.e. PV V1.0 LAI and NDVI derived from Sentinel-3 (S3) V1 TOC reflectance data. The yellow tones correspond to more density of points. The red lines define the upper and lower limits of the filtered training dataset.

2. Generation of the definition domain

The definition domain was used to check the input status when applying the neural network. It is defined by the convex hull made by the inputs in the training database. The convex hull is approximated by a hypercube with the same dimensions as those of the inputs of the neural network. Each dimension, corresponding to a specific spectral band, varies within the extremes defined by the minimum and maximum values for the bands (independently on the observational configuration) in the training database by allowing an absolute and relative tolerance of 0.01 (Table 10). The definition domain for Sentinel-3 neural networks was gridded using 5^{12} cells by dividing each dimension (the 12 selected OLCI spectral bands) into 5 equally spaced steps between the extrema (Figure 36). The definition domain for PROBA-V neural networks was gridded using 30^3 cells by dividing each dimension (the 3 selected PROBA-V spectral bands) into 30 equally spaced steps between the extrema (Figure 37). The definition domain was finally defined by the cells containing input data in the training database. Only a small fraction of the bounding cube is actually occupied by valid classes (classes where input reflectance values were actually observed in the training database).

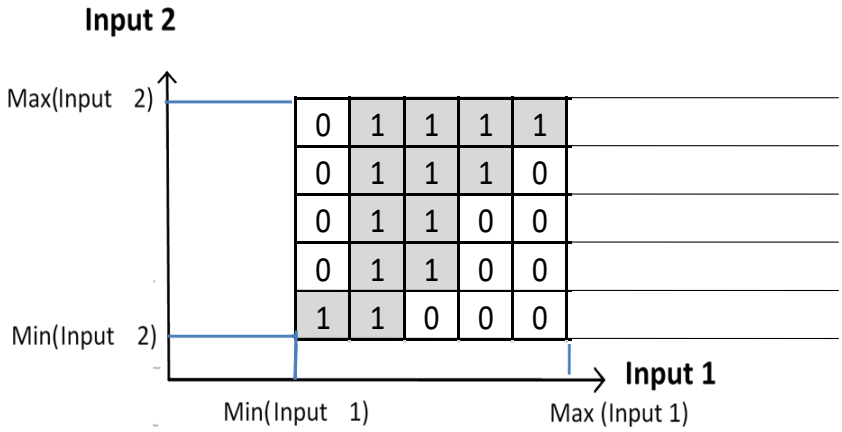


Figure 36: Schematic representation of the convex hull in the case of 2 dimensional inputs of the network. The convex hull is approximated by a regular grid. The cells with ones (the gray cells) correspond to cases of possible input range (i. e. inside the convex hull) while cells with zeros correspond to inputs out of range (outside the convex hull).

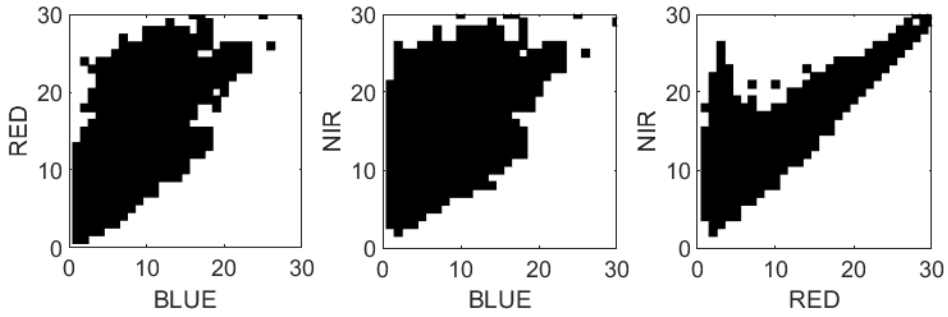


Figure 37: The convex hull corresponding to the definition domain of PROBA-V neural networks. Pixel will be declared as valid if they are within the area defined by the black areas. The 30 cells are distributed equally over the different reflectance ranges.

3. Neural network architecture

The neural networks architecture used in Version 2.0 is exactly the same as the one proposed for the generation of Version 1.0 of 300m LAI, FAPAR and FCover products [ImagineS_RP2.1_ATBD-LAI300m_I1.73]. The neural networks (Figure 38) are composed of:

- One input layer made of the input TOC data in the selected spectral bands (3 bands for PROBA-V and 12 bands for OLCI Sentinel-3) and the cosine of the 3 associated sun-view angles.

- One hidden layer with 5 neurons having tangent sigmoid transfer functions (Verger et al. 2011a).
- One output layer with a linear transfer function (Verger et al. 2011a).

The same architecture is used for PROBA-V and Sentinel-3 Version 2.0 neural networks. The single difference is the dimension of the input layer that for Sentinel-3 is 15 while it is 6 for PROBA-V since more spectral bands are available for OLCI Sentinel-3. This neural network architecture is in good agreement with what was proposed by Verger et al. (2011a).

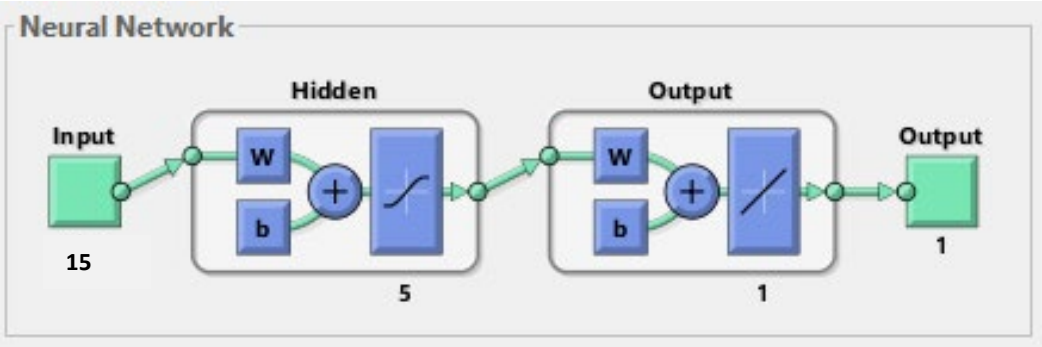


Figure 38: Neural network architecture for the estimation of the considered biophysical variables. It is made of 1 hidden layer with 5 neurons having sigmoidal transfer functions, and one output layer with one linear neuron. The dimension of the input layer is 15 for Sentinel-3 and 6 for PROBA-V.

4. Learning process

The learning process is mainly made of two elements: the training database that was described previously, and the learning rule that is now described. The Levenberg–Marquardt optimization algorithm is used to adjust the synaptic weights and neuron bias to get the best agreement between the output simulated by the network and the corresponding value of canopy biophysical variable simulated in the training database. The initial values of the weights and biases were set to a random value between -1.0 and +1.0. To prevent from hyper-specialization, a subset of the training database is used to control whether the network starts to hyper-specialize, i. e. represents the particular features of the training data set and therefore losing its capacity to describe the general features to be extracted. When this happens, the optimization process is stopped. The available data in the training dataset (n=55714 for Sentinel-3 and n=57810 for PROBA-V) were randomly split into a learning dataset made of 70% of the data available, while the 30% remaining part (test dataset) was used for hyper-specialization and theoretical evaluation (Verger et al. 2011a).

Evergreen Broadleaf Forests (EBFs) are characterized by specific behaviors as compared to other biomes: (i) although they are characterized by high variables' values, the near infrared reflectance

level is lower than for the other biomes for a given value of the variables (Asner, 1998). This is mainly due to thicker leaves (Jacquemoud and Baret, 1990). And, (ii) the temporal profiles of input TOC are characterized by a high level of noise while their values are expected to remain relatively stable during the whole year with low seasonality (Verger et al., 2015) (Figure 16, top). This is mainly due to the remaining atmospheric effects and cloud contamination since EBFs are located in latitudes with a high occurrence of clouds. To avoid the interference of these factors on the training for the other biomes, it was decided to consider two different processing chains whether the biome is an EBF or not.

The training data sets were different for EBF and nonEBF. For nonEBF, we discarded all the BELMANIP2.1 sites that were identified as EBF by GEOCLIM (Verger et al., 2015) to perform better estimates on high reflectance for non EBF biomes. Conversely, for EBF, we did not use EBF BELMANIP2.1 sites solely but the whole set of BELMANIP2.1 sites to be able to get valuable results in case the EBF identification is erroneous. For the nonEBF case, a biome distinction based on an external land cover is not used to avoid possible misclassification issues.

Ten networks were trained in parallel to retrieve the canopy biophysical variables, each corresponding to independent random drawing of the initial values of the synaptic weights and bias. Finally, the network kept for implementation is the one that provided the best performances in terms of RMSE, coefficient of determination (R^2), slope and offset of the linear regression over the test data set.

Figure 39 and Figure 40 show the theoretical performances of the EBF and nonEBF neural networks selected to estimate V2.0 LAI, FAPAR and FCover products from PROBA-V and Sentinel-3 data, respectively, as evaluated over the test dataset. The training process was quite successful in all cases with slightly better performances for PROBA-V neural networks because the same sensor was used for the input and output data: high correlation close to the line 1:1, and RMSE lower than 0.2 LAI, 0.03 FAPAR and 0.04 FCover. The slopes of the linear regression are close to 1.

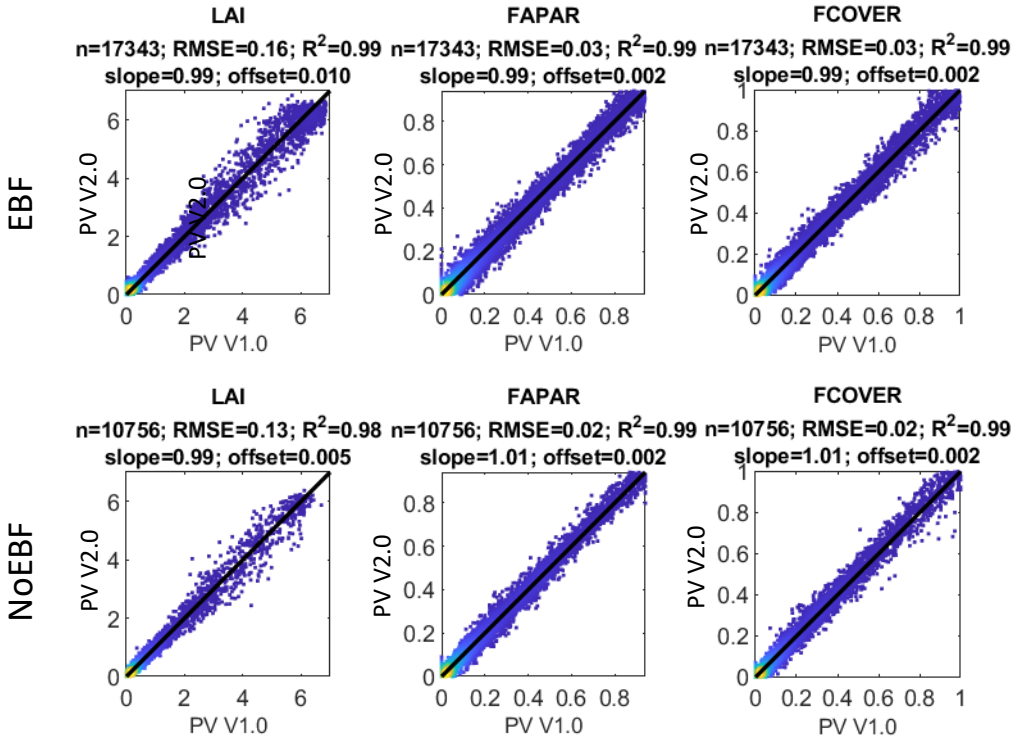


Figure 39: Theoretical performances of the neural networks used for the estimation of PROBA-V LAI, FAPAR and FCover V2.0 products. Top: EBF, Bottom: Non EBF. Neural network V2.0 PROBA-V (PV) predicted outputs and the V1.0 PROBA-V target products in the validation dataset are displayed as a density plot. The yellow tones correspond to more density of points. The continuous line is the 1:1 line. The number of samples (n), the root mean square error (RMSE), the coefficient of determination (R^2), the slope and the intercept of the linear regression are indicated.

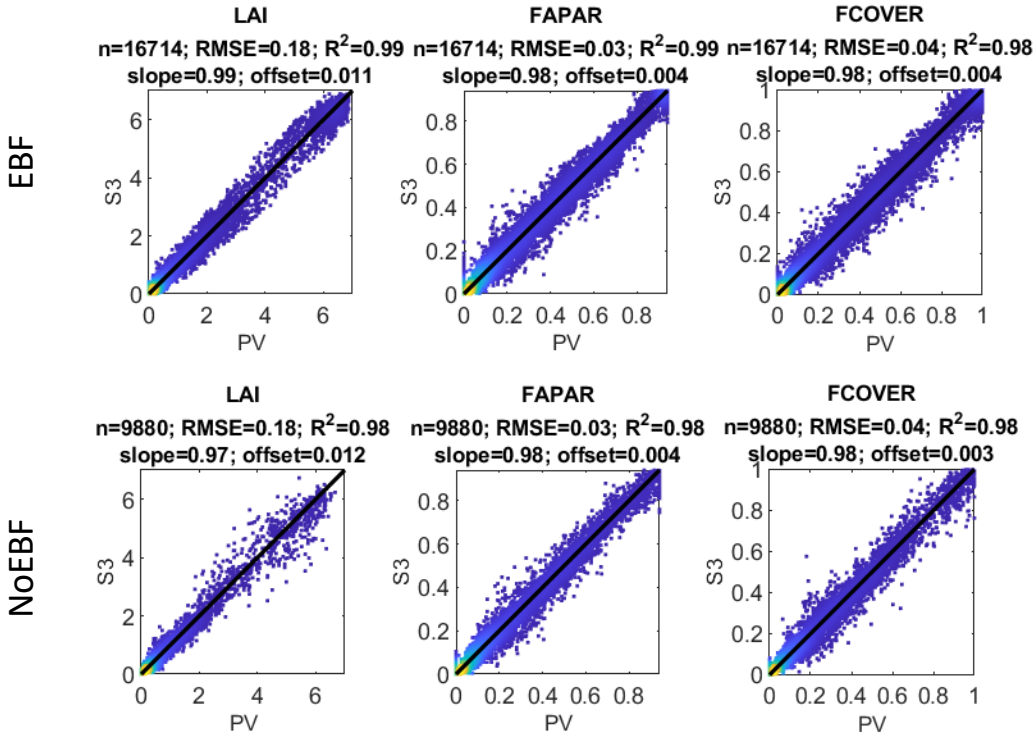


Figure 40: Theoretical performances of the neural networks used for the estimation of Sentinel-3 LAI, FAPAR and FCover V2.0 products. Top: EBF, Bottom: Non EBF. Neural network V2.0 Sentinel-3 (S3) predicted outputs and the V1.0 PROBA-V (PV) target products in the validation dataset are displayed as a density plot. The yellow tones correspond to more density of points. The continuous line is the 1:1 line. The number of samples (n), the root mean square error (RMSE), the coefficient of determination (R^2), the slope and the intercept of the linear regression are indicated.